

Deep Residual Learning for Image Recognition

Kaiming He Xiangyu Zhang Shaoqing Ren Jian Sun
Microsoft Research
{kahe, v-xiangz, v-shren, jiansun}@microsoft.com

Abstract

Deeper neural networks are more difficult to train. We present a residual learning framework to ease the training of networks that are substantially deeper than those used previously. We explicitly reformulate the layers as learning residual functions with reference to the layer inputs, instead of learning unreferenced functions. We provide comprehensive empirical evidence showing that these residual networks are easier to optimize, and can gain accuracy from considerably increased depth. On the ImageNet dataset we evaluate residual nets with a depth of up to 152 layers—8× deeper than VGG nets [41] but still having lower complexity. An ensemble of these residual nets achieves 3.57% error on the ImageNet test set. This result won the 1st place on the ILSVRC 2015 classification task. We also present analysis on CIFAR-10 with 100 and 1000 layers.

The depth of representations is of central importance for many visual recognition tasks. Solely due to our extremely deep representations, we obtain a 28% relative improvement on the COCO object detection dataset. Deep residual nets are foundations of our submissions to ILSVRC & COCO 2015 competitions¹, where we also won the 1st places on the tasks of ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation.

1. Introduction

Deep convolutional neural networks [22, 21] have led to a series of breakthroughs for image classification [21, 50, 40]. Deep networks naturally integrate low/mid/high-level features [50] and classifiers in an end-to-end multi-layer fashion, and the “levels” of features can be enriched by the number of stacked layers (depth). Recent evidence [41, 44] reveals that network depth is of crucial importance, and the leading results [41, 44, 13, 16] on the challenging ImageNet dataset [36] all exploit “very deep” [41] models, with a depth of sixteen [41] to thirty [16]. Many other non-trivial visual recognition tasks [8, 12, 7, 32, 27] have also

¹<http://image-net.org/challenges/LSVRC/2015/> and <http://mscoco.org/dataset/#detections-challenge2015>.

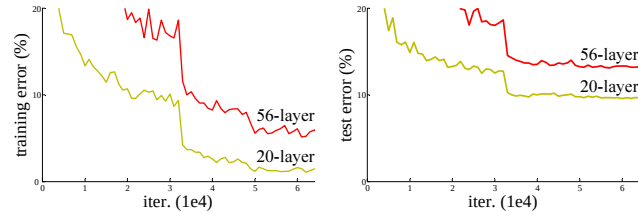


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

greatly benefited from very deep models.

Driven by the significance of depth, a question arises: *Is learning better networks as easy as stacking more layers?* An obstacle to answering this question was the notorious problem of vanishing/exploding gradients [1, 9], which hamper convergence from the beginning. This problem, however, has been largely addressed by normalized initialization [23, 9, 37, 13] and intermediate normalization layers [16], which enable networks with tens of layers to start converging for stochastic gradient descent (SGD) with back-propagation [22].

When deeper networks are able to start converging, a degradation problem has been exposed: with the network depth increasing, accuracy gets saturated (which might be unsurprising) and then degrades rapidly. Unexpectedly, such degradation is *not caused by overfitting*, and adding more layers to a suitably deep model leads to *higher training error*, as reported in [11, 42] and thoroughly verified by our experiments. Fig. 1 shows a typical example.

The degradation (of training accuracy) indicates that not all systems are similarly easy to optimize. Let us consider a shallower architecture and its deeper counterpart that adds more layers onto it. There exists a solution *by construction* to the deeper model: the added layers are *identity* mapping, and the other layers are copied from the learned shallower model. The existence of this constructed solution indicates that a deeper model should produce no higher training error than its shallower counterpart. But experiments show that our current solvers on hand are unable to find solutions that

用于图像识别的深度残差学习

何恺明 张祥雨 任少卿 孙剑 微软研究院{kahe, v-xiangz, v-shren, jiansun}@microsoft.com

摘要

更深的神经网络更难训练。我们提出了一个残差学习框架，以减轻对比以前所用网络深得多的网络的训练难度。我们明确地将层重新表述为参考层输入学习残差函数，而不是学习无参考的函数。我们提供了全面的经验证据，表明这些残差网络更容易优化，并且能够从大幅增加的深度中获得精度提升。在 ImageNet 数据集上，我们评估了深度高达 152 层的残差网络——8×比 VGG 网络更深，[41] 但复杂度仍然更低。这些残差网络的集成在 ImageNet 测试集上取得了 3.57% 的错误率。该结果在 ILSVRC 2015 分类任务中获得了第一名。我们还展示了对具有 100 层和 1000 层的 CIFAR-10 的分析。

表征的深度对于许多视觉识别任务至关重要。仅仅由于我们极深的表征，我们在 COCO 目标检测数据集上获得了 28% 的相对提升。深度残差网络是我们提交给 ILSVRC 和 COCO 2015 竞赛¹，的基础，在这些竞赛中，我们还在 ImageNet 检测、ImageNet 定位、COCO 检测和 COCO 分割任务上获得了第一名。

1. 引言

深度卷积神经网络 [22, 21] 在图像分类方面带来了一系列突破 [21, 50, 40]。深度网络以端到端的多层方式自然整合低/中/高层特征 [50] 与分类器，而特征的“层级”可通过堆叠层数（深度）得到丰富。近期证据[41, 44] 表明，网络深度至关重要，而领先结果 [41, 44, 13, 16] 在具有挑战性的 ImageNet 数据集 [36] 上全都采用了“非常深”的 [41] 模型，深度为十六 [41] 到三十 [16]。许多其他非平凡的视觉识别任务 [8, 12, 7, 32, 27] 也

¹<http://image-net.org/challenges/LSVRC/2015/> 和 <http://mscoco.org/dataset/#detections-challenge2015>。

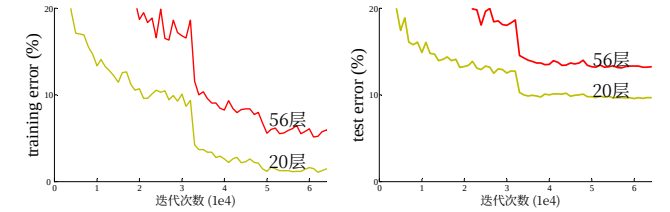


图 1. 在 CIFAR-10 上，20 层和 56 层“普通”网络的训练误差（左）和测试误差（右）。更深的网络具有更高的训练误差，因而测试误差也更高。ImageNet 上的类似现象如图 4 所示。

从非常深的模型中受益匪浅。

受深度重要性的驱动，一个问题随之产生：学习更好的网络是否像堆叠更多层一样简单？回答这个问题的一个障碍是臭名昭著的梯度消失/爆炸问题 [1, 9]，它从一开始就阻碍收敛。然而，这个问题已在很大程度上通过归一化初始化 [23, 9, 37, 13] 和中间归一化层[16]，得到解决，这些方法使具有数十层的网络能够开始使用带反向传播的随机梯度下降（SGD）进行收敛 [22]。

当更深的网络能够开始收敛时，一个退化问题已经暴露出来：随着网络深度增加，准确率会饱和（这也许并不令人意外），随后迅速下降。出乎意料的是，这种退化并非由过拟合导致，而向一个适当深的模型添加更多层会导致更高的训练误差，正如 [11, 42] 所报告并由我们的实验充分验证。图 1 展示了一个典型示例。

训练准确率的退化表明，并非所有系统都同样易于优化。让我们考虑一个较浅的架构，以及在其上添加更多层所得到的更深对应架构。对于更深模型，存在一个根据构造得到的解：新增的层是恒等映射，其他层则复制自学习到的较浅模型。这种构造解的存在表明，更深模型产生的训练误差不应高于其较浅对应模型。但实验表明，我们当前手头的求解器无法找到这样的解，这些解

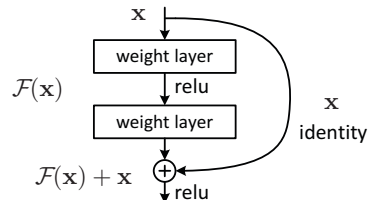


Figure 2. Residual learning: a building block.

are comparably good or better than the constructed solution (or unable to do so in feasible time).

In this paper, we address the degradation problem by introducing a *deep residual learning* framework. Instead of hoping each few stacked layers directly fit a desired underlying mapping, we explicitly let these layers fit a residual mapping. Formally, denoting the desired underlying mapping as $\mathcal{H}(\mathbf{x})$, we let the stacked nonlinear layers fit another mapping of $\mathcal{F}(\mathbf{x}) := \mathcal{H}(\mathbf{x}) - \mathbf{x}$. The original mapping is recast into $\mathcal{F}(\mathbf{x}) + \mathbf{x}$. We hypothesize that it is easier to optimize the residual mapping than to optimize the original, unreferenced mapping. To the extreme, if an identity mapping were optimal, it would be easier to push the residual to zero than to fit an identity mapping by a stack of nonlinear layers.

The formulation of $\mathcal{F}(\mathbf{x}) + \mathbf{x}$ can be realized by feedforward neural networks with “shortcut connections” (Fig. 2). Shortcut connections [2, 34, 49] are those skipping one or more layers. In our case, the shortcut connections simply perform *identity* mapping, and their outputs are added to the outputs of the stacked layers (Fig. 2). Identity shortcut connections add neither extra parameter nor computational complexity. The entire network can still be trained end-to-end by SGD with backpropagation, and can be easily implemented using common libraries (*e.g.*, Caffe [19]) without modifying the solvers.

We present comprehensive experiments on ImageNet [36] to show the degradation problem and evaluate our method. We show that: 1) Our extremely deep residual nets are easy to optimize, but the counterpart “plain” nets (that simply stack layers) exhibit higher training error when the depth increases; 2) Our deep residual nets can easily enjoy accuracy gains from greatly increased depth, producing results substantially better than previous networks.

Similar phenomena are also shown on the CIFAR-10 set [20], suggesting that the optimization difficulties and the effects of our method are not just akin to a particular dataset. We present successfully trained models on this dataset with over 100 layers, and explore models with over 1000 layers.

On the ImageNet classification dataset [36], we obtain excellent results by extremely deep residual nets. Our 152-layer residual net is the deepest network ever presented on ImageNet, while still having lower complexity than VGG nets [41]. Our ensemble has **3.57%** top-5 error on the

ImageNet *test* set, and won the *1st place in the ILSVRC 2015 classification competition*. The extremely deep representations also have excellent generalization performance on other recognition tasks, and lead us to further *win the 1st places on: ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation* in ILSVRC & COCO 2015 competitions. This strong evidence shows that the residual learning principle is generic, and we expect that it is applicable in other vision and non-vision problems.

2. Related Work

Residual Representations. In image recognition, VLAD [18] is a representation that encodes by the residual vectors with respect to a dictionary, and Fisher Vector [30] can be formulated as a probabilistic version [18] of VLAD. Both of them are powerful shallow representations for image retrieval and classification [4, 48]. For vector quantization, encoding residual vectors [17] is shown to be more effective than encoding original vectors.

In low-level vision and computer graphics, for solving Partial Differential Equations (PDEs), the widely used Multigrid method [3] reformulates the system as subproblems at multiple scales, where each subproblem is responsible for the residual solution between a coarser and a finer scale. An alternative to Multigrid is hierarchical basis preconditioning [45, 46], which relies on variables that represent residual vectors between two scales. It has been shown [3, 45, 46] that these solvers converge much faster than standard solvers that are unaware of the residual nature of the solutions. These methods suggest that a good reformulation or preconditioning can simplify the optimization.

Shortcut Connections. Practices and theories that lead to shortcut connections [2, 34, 49] have been studied for a long time. An early practice of training multi-layer perceptrons (MLPs) is to add a linear layer connected from the network input to the output [34, 49]. In [44, 24], a few intermediate layers are directly connected to auxiliary classifiers for addressing vanishing/exploding gradients. The papers of [39, 38, 31, 47] propose methods for centering layer responses, gradients, and propagated errors, implemented by shortcut connections. In [44], an “inception” layer is composed of a shortcut branch and a few deeper branches.

Concurrent with our work, “highway networks” [42, 43] present shortcut connections with gating functions [15]. These gates are data-dependent and have parameters, in contrast to our identity shortcuts that are parameter-free. When a gated shortcut is “closed” (approaching zero), the layers in highway networks represent *non-residual* functions. On the contrary, our formulation always learns residual functions; our identity shortcuts are never closed, and all information is always passed through, with additional residual functions to be learned. In addition, high-

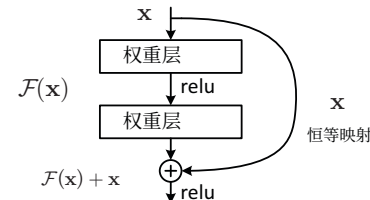


图 2. 残差学习：一个构建块。

与构造解相比效果相当或更好（或无法在可行时间内做到这一点）。

在本文中，我们通过引入深度残差学习框架来解决退化问题。我们不是寄希望于每几个堆叠层直接拟合一个期望的底层映射，而是显式地让这些层拟合一个残差映射。形式上，将期望的底层映射记为 $\mathcal{H}(\mathbf{x})$ ，我们让堆叠的非线性层拟合另一个映射 $\mathcal{F}(\mathbf{x}) := \mathcal{H}(\mathbf{x}) - \mathbf{x}$ 。原始映射被重写为 $\mathcal{F}(\mathbf{x}) + \mathbf{x}$ 。我们假设，优化残差映射比优化原始的、无参照的映射更容易。极端情况下，如果恒等映射是最优的，那么将残差推向零要比通过堆叠的非线性层来拟合恒等映射更容易。

$\mathcal{F}(\mathbf{x}) + \mathbf{x}$ 的公式可以通过带有“捷径连接”的前馈神经网络实现（图2）。捷径连接 [2, 34, 49] 是指跳过一层或多层的连接。在我们的情况中，捷径连接仅执行 恒等 映射，其输出被加到堆叠层的输出上（图2）。恒等捷径连接既不增加额外参数，也不增加计算复杂度。整个网络仍然可以通过带有反向传播的SGD进行端到端训练，并且可以使用常见库（例如，Caffe [19]）轻松实现，而无需修改求解器。

我们在ImageNet[36] 上进行了全面的实验，以展示退化问题并评估我们的方法。我们表明：1) 我们的极深残差网络易于优化，但对应的“普通”网络（仅简单堆叠层）在深度增加时表现出更高的训练误差；2) 我们的深度残差网络可以轻松地大幅增加的深度中获得精度提升，产生的结果显著优于之前的网络。

类似现象也在 CIFAR-10 数据集上出现[20]，，这表明优化困难以及我们方法的效果并非只与某一特定数据集相似。我们展示了在该数据集上成功训练的超过 100 层的模型，并探索了超过 1000 层的模型。

在 ImageNet 分类数据集上， [36], 我们通过极深的残差网络获得了出色的结果。我们的 152 层残差网络是 ImageNet 上有史以来提出的最深网络，同时复杂度仍低于 VGG 网络 [41]。我们的集成模型有 **3.57%** top-5 错误率，在

ImageNet 测试集，并赢得了 *ILSVRC 2015* 分类竞赛的第一名。极深的表示在其他识别任务上也具有出色的泛化性能，并使我们进一步在 ILSVRC & COCO 2015 竞赛中赢得以下项目的第一名：*ImageNet* 检测、*ImageNet* 定位、*COCO* 检测和 *COCO* 分割。这一有力证据表明，残差学习原理具有通用性，我们预期它适用于其他视觉和非视觉问题。

2. 相关工作

残差表示。 在图像识别中，VLAD[18]是一种通过相对于字典的残差向量进行编码的表示，而 Fisher 向量 [30] 可被形式化为 VLAD 的概率版本 [18]。两者都是用于图像检索和分类的强大浅层表示 [4, 48]。对于向量量化，编码残差向量 [17] 被证明比编码原始向量更有效。

在底层视觉和计算机图形学中，为求解偏微分方程（PDEs），广泛使用的多重网格方法 [3] 将系统重新表述为多个尺度上的子问题，其中每个子问题负责较粗尺度与较细尺度之间的残差解。多重网格的一种替代方法是分层基预处理 [45, 46]，，它依赖于表示两个尺度之间残差向量的变量。已有研究表明 [3, 45, 46]，这些求解器比那些未意识到解的残差性质的标准求解器收敛得快得多。这些方法表明，良好的重新表述或预处理可以简化优化。

快捷连接。 导致快捷连接的实践与理论 [2, 34, 49] 已经被研究了很长时间。训练多层感知机（MLP）的一种早期做法是添加一个从网络输入连接到输出的线性层 [34, 49]。在 [44, 24]，中，若干中间层直接连接到辅助分类器，以应对梯度消失/爆炸问题。文献 [39, 38, 31, 47]提出了对层响应、梯度和传播误差进行中心化的方法，这些方法由快捷连接实现。在 [44]，中，一个“inception”层由快捷分支和若干更深的分支组成。

与我们工作同期，“高速网络” [42, 43]提出带有门控函数的快捷连接 [15]。这些门依赖数据并带有参数，而我们的恒等快捷连接则无参数。当门控快捷连接“关闭”（趋近于零）时，高速网络中的层表示 非残差 函数。相反，我们的形式始终学习残差函数；我们的恒等快捷连接从不关闭，所有信息始终会传递过去，同时还需要学习额外的残差函数。此外，高

way networks have not demonstrated accuracy gains with extremely increased depth (*e.g.*, over 100 layers).

3. Deep Residual Learning

3.1. Residual Learning

Let us consider $\mathcal{H}(\mathbf{x})$ as an underlying mapping to be fit by a few stacked layers (not necessarily the entire net), with $\mathbf{x}$ denoting the inputs to the first of these layers. If one hypothesizes that multiple nonlinear layers can asymptotically approximate complicated functions², then it is equivalent to hypothesize that they can asymptotically approximate the residual functions, *i.e.*, $\mathcal{H}(\mathbf{x}) - \mathbf{x}$ (assuming that the input and output are of the same dimensions). So rather than expect stacked layers to approximate $\mathcal{H}(\mathbf{x})$, we explicitly let these layers approximate a residual function $\mathcal{F}(\mathbf{x}) := \mathcal{H}(\mathbf{x}) - \mathbf{x}$. The original function thus becomes $\mathcal{F}(\mathbf{x}) + \mathbf{x}$. Although both forms should be able to asymptotically approximate the desired functions (as hypothesized), the ease of learning might be different.

This reformulation is motivated by the counterintuitive phenomena about the degradation problem (Fig. 1, left). As we discussed in the introduction, if the added layers can be constructed as identity mappings, a deeper model should have training error no greater than its shallower counterpart. The degradation problem suggests that the solvers might have difficulties in approximating identity mappings by multiple nonlinear layers. With the residual learning reformulation, if identity mappings are optimal, the solvers may simply drive the weights of the multiple nonlinear layers toward zero to approach identity mappings.

In real cases, it is unlikely that identity mappings are optimal, but our reformulation may help to precondition the problem. If the optimal function is closer to an identity mapping than to a zero mapping, it should be easier for the solver to find the perturbations with reference to an identity mapping, than to learn the function as a new one. We show by experiments (Fig. 7) that the learned residual functions in general have small responses, suggesting that identity mappings provide reasonable preconditioning.

3.2. Identity Mapping by Shortcuts

We adopt residual learning to every few stacked layers. A building block is shown in Fig. 2. Formally, in this paper we consider a building block defined as:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}. \quad (1)$$

Here $\mathbf{x}$ and $\mathbf{y}$ are the input and output vectors of the layers considered. The function $\mathcal{F}(\mathbf{x}, \{W_i\})$ represents the residual mapping to be learned. For the example in Fig. 2 that has two layers, $\mathcal{F} = W_2\sigma(W_1\mathbf{x})$ in which σ denotes

ReLU [29] and the biases are omitted for simplifying notations. The operation $\mathcal{F} + \mathbf{x}$ is performed by a shortcut connection and element-wise addition. We adopt the second nonlinearity after the addition (*i.e.*, $\sigma(\mathbf{y})$, see Fig. 2).

The shortcut connections in Eqn.(1) introduce neither extra parameter nor computation complexity. This is not only attractive in practice but also important in our comparisons between plain and residual networks. We can fairly compare plain/residual networks that simultaneously have the same number of parameters, depth, width, and computational cost (except for the negligible element-wise addition).

The dimensions of $\mathbf{x}$ and $\mathcal{F}$ must be equal in Eqn.(1). If this is not the case (*e.g.*, when changing the input/output channels), we can perform a linear projection W_s by the shortcut connections to match the dimensions:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + W_s\mathbf{x}. \quad (2)$$

We can also use a square matrix W_s in Eqn.(1). But we will show by experiments that the identity mapping is sufficient for addressing the degradation problem and is economical, and thus W_s is only used when matching dimensions.

The form of the residual function $\mathcal{F}$ is flexible. Experiments in this paper involve a function $\mathcal{F}$ that has two or three layers (Fig. 5), while more layers are possible. But if $\mathcal{F}$ has only a single layer, Eqn.(1) is similar to a linear layer: $\mathbf{y} = W_1\mathbf{x} + \mathbf{x}$, for which we have not observed advantages.

We also note that although the above notations are about fully-connected layers for simplicity, they are applicable to convolutional layers. The function $\mathcal{F}(\mathbf{x}, \{W_i\})$ can represent multiple convolutional layers. The element-wise addition is performed on two feature maps, channel by channel.

3.3. Network Architectures

We have tested various plain/residual nets, and have observed consistent phenomena. To provide instances for discussion, we describe two models for ImageNet as follows.

Plain Network. Our plain baselines (Fig. 3, middle) are mainly inspired by the philosophy of VGG nets [41] (Fig. 3, left). The convolutional layers mostly have 3×3 filters and follow two simple design rules: (i) for the same output feature map size, the layers have the same number of filters; and (ii) if the feature map size is halved, the number of filters is doubled so as to preserve the time complexity per layer. We perform downsampling directly by convolutional layers that have a stride of 2. The network ends with a global average pooling layer and a 1000-way fully-connected layer with softmax. The total number of weighted layers is 34 in Fig. 3 (middle).

It is worth noticing that our model has *fewer* filters and *lower* complexity than VGG nets [41] (Fig. 3, left). Our 34-layer baseline has 3.6 billion FLOPs (multiply-adds), which is only 18% of VGG-19 (19.6 billion FLOPs).

速网络尚未在深度极大增加（例如，超过 100 层）时展现出精度提升。

3. 深度残差学习

3.1. 残差学习

让我们考虑将 $\mathcal{H}(\mathbf{x})$ 视为要由若干堆叠层（不一定是整个网络）拟合的底层映射，其中 $\mathbf{x}$ 表示这些层中第一层的输入。如果假设多个非线性层可以渐近逼近复杂函数²，那么这就等价于假设它们可以渐近逼近残差函数，即， $\mathcal{H}(\mathbf{x}) - \mathbf{x}$ （假设输入和输出具有相同维度）。因此，与其期望堆叠层逼近 $\mathcal{H}(\mathbf{x})$ ，我们不如显式地让这些层逼近一个残差函数 $\mathcal{F}(\mathbf{x}) := \mathcal{H}(\mathbf{x}) - \mathbf{x}$ 。原始函数因此变为 $\mathcal{F}(\mathbf{x}) + \mathbf{x}$ 。尽管这两种形式都应能够（如假设的那样）渐近逼近所需函数，但学习的难易程度可能不同。

这种重新表述的动机来自关于退化问题（图1左）的反直觉现象。正如我们在引言中讨论的，如果新增的层可以构造为恒等映射，那么更深的模型训练误差不应大于其较浅的对应模型。退化问题表明，求解器可能难以通过多个非线性层逼近恒等映射。借助残差学习这一重新表述，如果恒等映射是最优的，求解器可以简单地将多个非线性层的权重推向零，以逼近恒等映射。

在实际情况下，恒等映射不太可能是最优的，但我们的重构可能有助于对问题进行预条件化。如果最优函数比零映射更接近恒等映射，那么求解器以恒等映射为参考来寻找扰动，应该比将该函数作为一个全新函数来学习更容易。我们通过实验（图 7）表明，学习到的残差函数通常响应较小，这说明恒等映射提供了合理的预条件。

3.2. 通过快捷连接实现恒等映射

我们对每几个堆叠层采用残差学习。图 2 展示了一个构建块。形式上，在本文中，我们考虑一个定义如下的构建块：

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x}. \quad (1)$$

这里 $\mathbf{x}$ 和 $\mathbf{y}$ 是所考虑层的输入和输出向量。函数 $\mathcal{F}(\mathbf{x}, \{W_i\})$ 表示要学习的残差映射。对于图 2 中具有两层的示例， $\mathcal{F} = W_2\sigma(W_1\mathbf{x})$ ，其中 σ 表示

²然而，这一假设仍是一个未决问题。参见 [28]。

ReLU [29]，并且为简化记号省略了偏置项。运算 $\mathcal{F} + \mathbf{x}$ 由快捷连接和逐元素加法完成。我们在加法之后采用第二个非线性（即， $\sigma(\mathbf{y})$ ，见图 2）。

公式 (1) 中的快捷连接既不引入额外参数，也不引入计算复杂度。这不仅在实践中具有吸引力，而且在我们对普通网络和残差网络进行比较时也很重要。我们可以公平地比较同时具有相同参数数量、深度、宽度和计算代价的普通/残差网络（除了可忽略的逐元素加法）。

$\mathbf{x}$ 和 $\mathcal{F}$ 的维度在式(1)中必须相等。如果不是这种情况（例如，在改变输入/输出通道时），我们可以通过快捷连接执行线性投影 W_s ，以匹配维度：

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + W_s\mathbf{x}. \quad (2)$$

我们也可以在式(1)中使用方阵 W_s 。但我们将通过实验表明，恒等映射足以解决退化问题，而且经济，因此 W_s 仅在匹配维度时使用。

残差函数 $\mathcal{F}$ 的形式是灵活的。本文中的实验涉及一个具有两层或三层（图 5）的函数 $\mathcal{F}$ ，但也可以有更多层。但如果 $\mathcal{F}$ 只有单层，式(1)就类似于一个线性层： $\mathbf{y} = W_1\mathbf{x} + \mathbf{x}$ ，对此我们尚未观察到优势。

我们还要指出，尽管为简单起见，上述记号是针对全连接层的，但它们同样适用于卷积层。函数 $\mathcal{F}(\mathbf{x}, \{W_i\})$ 可以表示多个卷积层。逐元素加法是在两个特征图上逐通道进行的。

3.3. 网络架构

我们测试了各种普通/残差网络，并观察到了一致的现象。为了提供讨论的实例，我们如下描述用于 ImageNet 的两个模型。

普通网络。 我们的普通基线（图 3，中）主要受到 VGG 网络 [41]（图 3，左）设计思想的启发。卷积层大多使用 3×3 滤波器，并遵循两条简单的设计规则：(i) 对于相同的输出特征图尺寸，各层具有相同数量的滤波器；(ii) 如果特征图尺寸减半，则滤波器数量加倍，以保持每层的时间复杂度。我们直接通过步长为 2 的卷积层进行下采样。网络以一个全局平均池化层和一个带有 softmax 的 1000 路全连接层结束。图 3（中）中带权重的层总数为 34。

值得注意的是，我们的模型具有比 VGG 网络 [41]（图3左）更少的滤波器和更低的复杂度。我们的34层基线具有36亿 FLOPs（乘加），仅为 VGG-19 的18%（196亿 FLOPs）。

²This hypothesis, however, is still an open question. See [28].

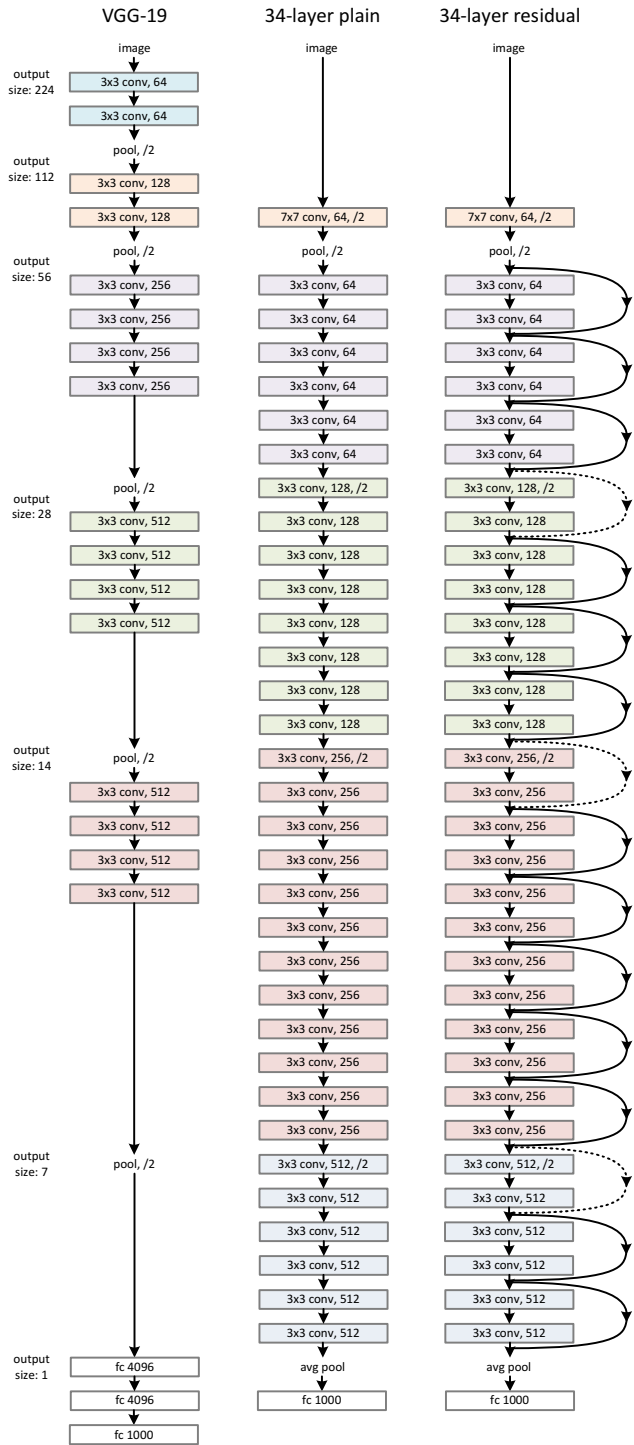


Figure 3. Example network architectures for ImageNet. **Left:** the VGG-19 model [41] (19.6 billion FLOPs) as a reference. **Middle:** a plain network with 34 parameter layers (3.6 billion FLOPs). **Right:** a residual network with 34 parameter layers (3.6 billion FLOPs). The dotted shortcuts increase dimensions. **Table 1** shows more details and other variants.

Residual Network. Based on the above plain network, we insert shortcut connections (Fig. 3, right) which turn the network into its counterpart residual version. The identity shortcuts (Eqn.(1)) can be directly used when the input and output are of the same dimensions (solid line shortcuts in Fig. 3). When the dimensions increase (dotted line shortcuts in Fig. 3), we consider two options: (A) The shortcut still performs identity mapping, with extra zero entries padded for increasing dimensions. This option introduces no extra parameter; (B) The projection shortcut in Eqn.(2) is used to match dimensions (done by 1×1 convolutions). For both options, when the shortcuts go across feature maps of two sizes, they are performed with a stride of 2.

3.4. Implementation

Our implementation for ImageNet follows the practice in [21, 41]. The image is resized with its shorter side randomly sampled in [256, 480] for scale augmentation [41]. A 224×224 crop is randomly sampled from an image or its horizontal flip, with the per-pixel mean subtracted [21]. The standard color augmentation in [21] is used. We adopt batch normalization (BN) [16] right after each convolution and before activation, following [16]. We initialize the weights as in [13] and train all plain/residual nets from scratch. We use SGD with a mini-batch size of 256. The learning rate starts from 0.1 and is divided by 10 when the error plateaus, and the models are trained for up to 60×10^4 iterations. We use a weight decay of 0.0001 and a momentum of 0.9. We do not use dropout [14], following the practice in [16].

In testing, for comparison studies we adopt the standard 10-crop testing [21]. For best results, we adopt the fully-convolutional form as in [41, 13], and average the scores at multiple scales (images are resized such that the shorter side is in $\{224, 256, 384, 480, 640\}$).

4. Experiments

4.1. ImageNet Classification

We evaluate our method on the ImageNet 2012 classification dataset [36] that consists of 1000 classes. The models are trained on the 1.28 million training images, and evaluated on the 50k validation images. We also obtain a final result on the 100k test images, reported by the test server. We evaluate both top-1 and top-5 error rates.

Plain Networks. We first evaluate 18-layer and 34-layer plain nets. The 34-layer plain net is in Fig. 3 (middle). The 18-layer plain net is of a similar form. See Table 1 for detailed architectures.

The results in Table 2 show that the deeper 34-layer plain net has higher validation error than the shallower 18-layer plain net. To reveal the reasons, in Fig. 4 (left) we compare their training/validation errors during the training procedure. We have observed the degradation problem - the

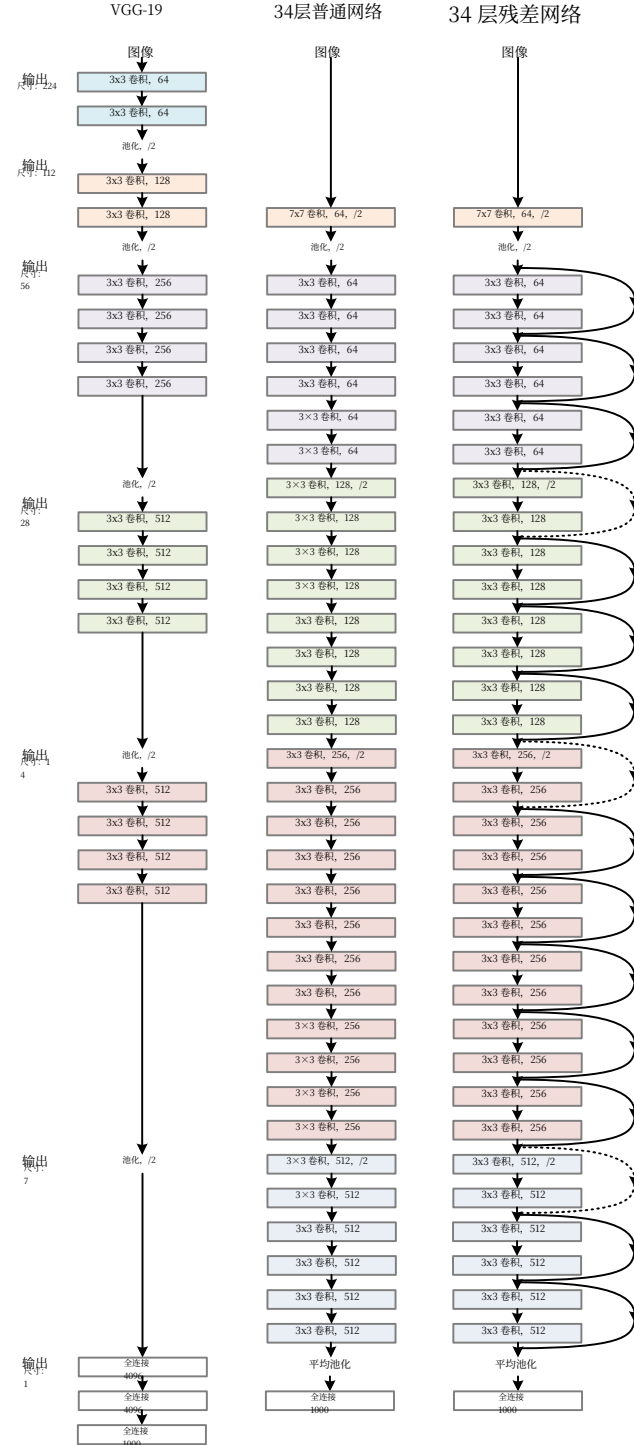


图 3. ImageNet 的示例网络架构。左：VGG-19 模型 [41]（196 亿 FLOPs）作为参考。中间：一个具有 34 个参数层的普通网络（36 亿 FLOPs）。右：一个具有 34 个参数层的残差网络（36 亿 FLOPs）。虚线捷径会增大维度。表 1 显示了更多细节和其他变体。

残差网络。 基于上述普通网络，我们插入快捷连接（图3右），将网络转换为对应的残差版本。当输入和输出具有相同维度时，可以直接使用恒等快捷连接（式(1)）（图3中的实线快捷连接）。当维度增加时（图3中的虚线快捷连接），我们考虑两种选项：(A) 快捷连接仍执行恒等映射，并为增加的维度填充额外的零项。此选项不引入额外参数；(B) 使用式(2)中的投影快捷连接来匹配维度（通过 1×1 卷积完成）。对于这两种选项，当快捷连接跨越两个尺寸的特征图时，它们以步长2执行。

3.4. 实现

我们针对 ImageNet 的实现遵循 [21, 41] 中的做法。图像被调整大小，其短边在 [256, 480] 中随机采样以进行尺度增强 [41]。从图像或其水平翻转中随机采样一个 224×224 裁剪区域，并减去逐像素均值 [21]。使用 [21] 中的标准颜色增强。我们在每个卷积之后、激活之前立即采用批归一化 (BN) [16]，遵循 [16]。我们按照 [13] 初始化权重，并从零开始训练所有普通 / 残差网络。我们使用小批量大小为 256 的 SGD。学习率从 0.1 开始，当误差趋于平稳时除以 10，模型训练最多 60×10^4 次迭代。我们使用 0.0001 的权重衰减和 0.9 的动量。我们不使用 dropout [14]，遵循 [16] 中的做法。

在测试中，对于比较研究，我们采用标准的 10 次裁剪测试 [21]。为获得最佳结果，我们采用如 [41, 13] 中的全卷积形式，并在多个尺度上对分数取平均（图像会被缩放，使短边位于 $\{224, 256, 384, 480, 640\}$ 中）。

4. 实验

4.1. ImageNet 分类

我们在由 1000 个类别组成的 ImageNet 2012 分类数据集 [36] 上评估我们的方法。模型在 128 万张训练图像上训练，并在 5 万张验证图像上评估。我们还获得了 10 万张测试图像上的最终结果，由测试服务器报告。我们同时评估 top-1 和 top-5 错误率。

普通网络。 我们首先评估 18 层和 34 层的普通网络。34 层普通网络如图 3（中）所示。18 层普通网络具有类似的形式。详细架构见表 1。

表2中的结果表明，更深的34层普通网络比较浅的18层普通网络具有更高的验证误差。为揭示原因，我们在图4（左）中比较了它们在训练过程中的训练/验证误差。我们观察到了退化问题——该

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
		3×3 max pool, stride 2				
conv2_x	56×56	$\begin{bmatrix} 3\times 3, 64 \\ 3\times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 64 \\ 3\times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3\times 3, 128 \\ 3\times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 128 \\ 3\times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3\times 3, 256 \\ 3\times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 256 \\ 3\times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3\times 3, 512 \\ 3\times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 512 \\ 3\times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

Table 1. Architectures for ImageNet. Building blocks are shown in brackets (see also Fig. 5), with the numbers of blocks stacked. Down-sampling is performed by conv3_1, conv4_1, and conv5_1 with a stride of 2.

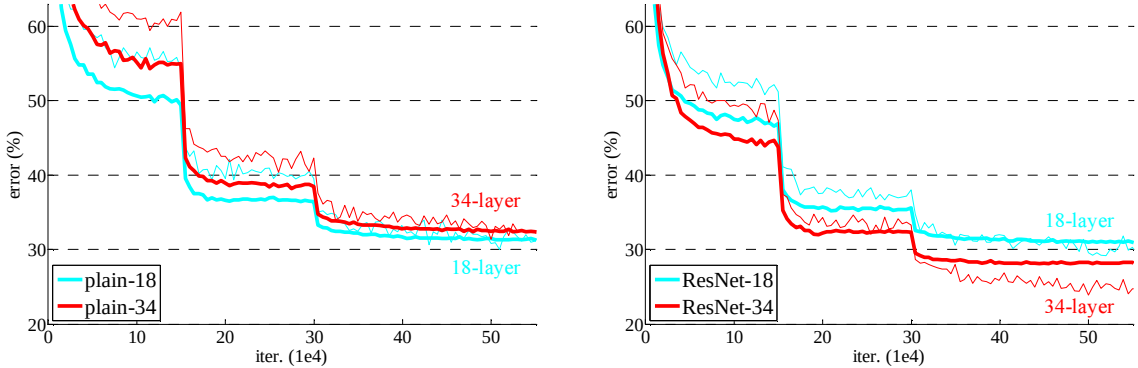


Figure 4. Training on **ImageNet**. Thin curves denote training error, and bold curves denote validation error of the center crops. Left: plain networks of 18 and 34 layers. Right: ResNets of 18 and 34 layers. In this plot, the residual networks have no extra parameter compared to their plain counterparts.

	plain	ResNet
18 layers	27.94	27.88
34 layers	28.54	25.03

Table 2. Top-1 error (% , 10-crop testing) on ImageNet validation. Here the ResNets have no extra parameter compared to their plain counterparts. Fig. 4 shows the training procedures.

34-layer plain net has higher *training* error throughout the whole training procedure, even though the solution space of the 18-layer plain network is a subspace of that of the 34-layer one.

We argue that this optimization difficulty is *unlikely* to be caused by vanishing gradients. These plain networks are trained with BN [16], which ensures forward propagated signals to have non-zero variances. We also verify that the backward propagated gradients exhibit healthy norms with BN. So neither forward nor backward signals vanish. In fact, the 34-layer plain net is still able to achieve competitive accuracy (Table 3), suggesting that the solver works to some extent. We conjecture that the deep plain nets may have exponentially low convergence rates, which impact the

reducing of the training error³. The reason for such optimization difficulties will be studied in the future.

Residual Networks. Next we evaluate 18-layer and 34-layer residual nets (*ResNets*). The baseline architectures are the same as the above plain nets, expect that a shortcut connection is added to each pair of 3×3 filters as in Fig. 3 (right). In the first comparison (Table 2 and Fig. 4 right), we use identity mapping for all shortcuts and zero-padding for increasing dimensions (option A). So they have *no extra parameter* compared to the plain counterparts.

We have three major observations from Table 2 and Fig. 4. First, the situation is reversed with residual learning – the 34-layer ResNet is better than the 18-layer ResNet (by 2.8%). More importantly, the 34-layer ResNet exhibits considerably lower training error and is generalizable to the validation data. This indicates that the degradation problem is well addressed in this setting and we manage to obtain accuracy gains from increased depth.

Second, compared to its plain counterpart, the 34-layer

³We have experimented with more training iterations (3×) and still observed the degradation problem, suggesting that this problem cannot be feasibly addressed by simply using more iterations.

层名称 输出尺寸	18层	34层	50层	101层	152层
conv1	112×112	7×7, 64, 步幅 2			
		3×3最大池化, 步长 2			
conv2_x	56×56	$\begin{bmatrix} 3\times 3, 64 \\ 3\times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 64 \\ 3\times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 64 \\ 3\times 3, 64 \\ 1\times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3\times 3, 128 \\ 3\times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 128 \\ 3\times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1\times 1, 128 \\ 3\times 3, 128 \\ 1\times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3\times 3, 256 \\ 3\times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 256 \\ 3\times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1\times 1, 256 \\ 3\times 3, 256 \\ 1\times 1, 1024 \end{bmatrix} \times 23$
conv5_x	7×7	$\begin{bmatrix} 3\times 3, 512 \\ 3\times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3\times 3, 512 \\ 3\times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1\times 1, 512 \\ 3\times 3, 512 \\ 1\times 1, 2048 \end{bmatrix} \times 3$
	1×1	平均池化, 1000维全连接层, softmax			
浮点运算次数		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9

表 1. 用于 ImageNet 的架构。构建模块在括号中显示（另见图 5），并给出了堆叠的模块数量。下采样由 conv3_1、conv4_1 和 conv5_1 以步幅 2 执行。

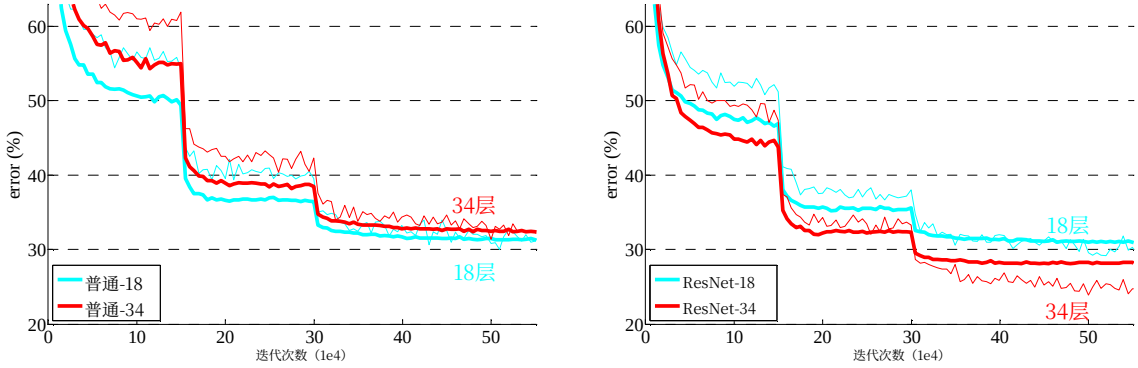


图 4. 在 **ImageNet** 上训练。细曲线表示训练误差，粗曲线表示中心裁剪的验证误差。左：18 层和 34 层的普通网络。右：18 层和 34 层的 ResNet。在此图中，残差网络与对应的普通网络相比没有额外参数。

	普通网络	ResNet
18 层	27.94	27.88
34 层	28.54	25.03

表 2. ImageNet 验证集上的 Top-1 误差（%，10-crop 测试）。此处 ResNets 与其普通对应网络相比没有额外参数。图 4 展示了训练过程。

34层普通网络在整个训练过程中都具有更高的训练误差，即使18层普通网络的解空间是34层普通网络解空间的一个子空间。

我们论证，这一优化困难不太可能 是由梯度消失引起的。这些普通网络使用 BN [16], 训练，这确保前向传播信号具有非零方差。我们还验证了在 BN 下反向传播的梯度表现出良好的范数。因此，前向和反向信号都没有消失。事实上，34 层普通网络仍能取得有竞争力的准确率（表 3），这表明求解器在一定程度上能够工作。我们推测，深层普通网络可能具有指数级低的收敛速率，从而影响

训练误差的降低³。此类优化困难的原因将在未来进行研究。

残差网络。 接下来我们评估 18 层和 34层残差网络（*ResNets*）。基线架构与上述普通网络相同，不同之处在于，如图 3（右）所示，每一对 3×3 滤波器都添加了捷径连接。在第一次比较中（表 2 和图 4 右侧），我们为所有捷径使用恒等映射，并为增加维度使用零填充（选项 A）。因此，与普通对应网络相比，它们没有额外参数。

从表 2 和图 4 中，我们有三项主要观察。首先，使用残差学习后情况发生了逆转——34 层 ResNet 优于 18 层 ResNet（高出 2.8%）。更重要的是，34 层 ResNet 表现出显著更低的训练误差，并且能够泛化到验证数据。这表明在这种设置下，退化问题得到了很好的解决，我们得以从增加的深度中获得精度提升。

其次，与其普通对应网络相比，34 层的

³我们尝试了更多训练迭代（3×），却依然观察到退化问题，这表明仅靠使用更多迭代无法切实解决该问题。

model	top-1 err.	top-5 err.
VGG-16 [41]	28.07	9.33
GoogLeNet [44]	-	9.15
PReLU-net [13]	24.27	7.38
plain-34	28.54	10.02
ResNet-34 A	25.03	7.76
ResNet-34 B	24.52	7.46
ResNet-34 C	24.19	7.40
ResNet-50	22.85	6.71
ResNet-101	21.75	6.05
ResNet-152	21.43	5.71

Table 3. Error rates (% , **10-crop** testing) on ImageNet validation. VGG-16 is based on our test. ResNet-50/101/152 are of option B that only uses projections for increasing dimensions.

method	top-1 err.	top-5 err.
VGG [41] (ILSVRC’ 14)	-	8.43 [†]
GoogLeNet [44] (ILSVRC’ 14)	-	7.89
VGG [41] (v5)	24.4	7.1
PReLU-net [13]	21.59	5.71
BN-inception [16]	21.99	5.81
ResNet-34 B	21.84	5.71
ResNet-34 C	21.53	5.60
ResNet-50	20.74	5.25
ResNet-101	19.87	4.60
ResNet-152	19.38	4.49

Table 4. Error rates (%) of **single-model** results on the ImageNet validation set (except [†] reported on the test set).

method	top-5 err. (test)
VGG [41] (ILSVRC’ 14)	7.32
GoogLeNet [44] (ILSVRC’ 14)	6.66
VGG [41] (v5)	6.8
PReLU-net [13]	4.94
BN-inception [16]	4.82
ResNet (ILSVRC’15)	3.57

Table 5. Error rates (%) of **ensembles**. The top-5 error is on the test set of ImageNet and reported by the test server.

ResNet reduces the top-1 error by 3.5% (Table 2), resulting from the successfully reduced training error (Fig. 4 right *vs.* left). This comparison verifies the effectiveness of residual learning on extremely deep systems.

Last, we also note that the 18-layer plain/residual nets are comparably accurate (Table 2), but the 18-layer ResNet converges faster (Fig. 4 right *vs.* left). When the net is “not overly deep” (18 layers here), the current SGD solver is still able to find good solutions to the plain net. In this case, the ResNet eases the optimization by providing faster convergence at the early stage.

Identity vs. Projection Shortcuts. We have shown that

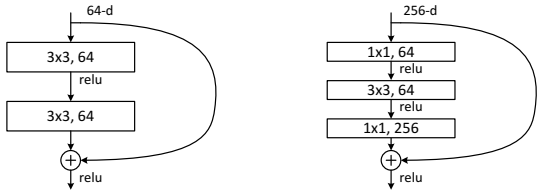


Figure 5. A deeper residual function $\mathcal{F}$ for ImageNet. Left: a building block (on 56×56 feature maps) as in Fig. 3 for ResNet-34. Right: a “bottleneck” building block for ResNet-50/101/152.

parameter-free, identity shortcuts help with training. Next we investigate projection shortcuts (Eqn.(2)). In Table 3 we compare three options: (A) zero-padding shortcuts are used for increasing dimensions, and all shortcuts are parameter-free (the same as Table 2 and Fig. 4 right); (B) projection shortcuts are used for increasing dimensions, and other shortcuts are identity; and (C) all shortcuts are projections.

Table 3 shows that all three options are considerably better than the plain counterpart. B is slightly better than A. We argue that this is because the zero-padded dimensions in A indeed have no residual learning. C is marginally better than B, and we attribute this to the extra parameters introduced by many (thirteen) projection shortcuts. But the small differences among A/B/C indicate that projection shortcuts are not essential for addressing the degradation problem. So we do not use option C in the rest of this paper, to reduce memory/time complexity and model sizes. Identity shortcuts are particularly important for not increasing the complexity of the bottleneck architectures that are introduced below.

Deeper Bottleneck Architectures. Next we describe our deeper nets for ImageNet. Because of concerns on the training time that we can afford, we modify the building block as a *bottleneck* design⁴. For each residual function $\mathcal{F}$, we use a stack of 3 layers instead of 2 (Fig. 5). The three layers are 1×1 , 3×3 , and 1×1 convolutions, where the 1×1 layers are responsible for reducing and then increasing (restoring) dimensions, leaving the 3×3 layer a bottleneck with smaller input/output dimensions. Fig. 5 shows an example, where both designs have similar time complexity.

The parameter-free identity shortcuts are particularly important for the bottleneck architectures. If the identity shortcut in Fig. 5 (right) is replaced with projection, one can show that the time complexity and model size are doubled, as the shortcut is connected to the two high-dimensional ends. So identity shortcuts lead to more efficient models for the bottleneck designs.

50-layer ResNet: We replace each 2-layer block in the

⁴Deeper *non*-bottleneck ResNets (*e.g.*, Fig. 5 left) also gain accuracy from increased depth (as shown on CIFAR-10), but are not as economical as the bottleneck ResNets. So the usage of bottleneck designs is mainly due to practical considerations. We further note that the degradation problem of plain nets is also witnessed for the bottleneck designs.

模型	top-1 错误率	top-5 错误率
VGG-16 [41]	28.07	9.33
GoogLeNet [44]	-	9.15
PReLU-net [13]	24.27	7.38
plain-34	28.54	10.02
ResNet-34 A	25.03	7.76
ResNet-34 B	24.52	7.46
ResNet-34 C	24.19	7.40
ResNet-50	22.85	6.71
ResNet-101	21.75	6.05
ResNet-152	21.43	5.71

表 3. ImageNet 验证集上的错误率 (% , **10 裁剪** 测试) 。 VGG-16 基于我们的测试。 ResNet-50/101/152 为选项 B, 仅使用投影来增加维度。

方法	top-1 错误率	top-5 错误率
VGG [41] (ILSVRC’ 14)	-	8.43 [†]
GoogLeNet [44] (ILSVRC’ 14)	-	7.89
VGG [41] (v5)	24.4	7.1
PReLU-net [13]	21.59	5.71
BN-inception [16]	21.99	5.81
ResNet-34 B	21.84	5.71
ResNet-34 C	21.53	5.60
ResNet-50	20.74	5.25
ResNet-101	19.87	4.60
ResNet-152	19.38	4.49

表 4. ImageNet 验证集上**单模型** 结果的错误率 (%) （除 [†] 在测试集上报告的以外）。

方法	top-5 错误率 (测试)
VGG [41] (ILSVRC’ 14)	7.32
GoogLeNet [44] (ILSVRC’ 14)	6.66
VGG [41] (v5)	6.8
PReLU-net [13]	4.94
BN-inception [16]	4.82
ResNet (ILSVRC’ 15)	3.57

表 5. **集成**的错误率 (%) 。 top-5 错误率是在 ImageNet 的测试集上, 并由测试服务器报告的。

ResNet 将 top-1 错误率降低了 3.5%（表 2），这源于成功降低的训练误差（图 4 右与左）。这一对比验证了残差学习在极深系统上的有效性。

最后, 我们还注意到, 18 层的普通/残差网络在准确率上大致相当（表 2），但 18 层 ResNet 收敛得更快（图 4 右与左）。当网络“不算过深”（这里为 18 层）时, 当前的 SGD 求解器仍能够为普通网络找到良好的解。在这种情况下, ResNet 通过在早期阶段提供更快收敛, 使优化变得更容易。

恒等映射与投影捷径。我们已经表明

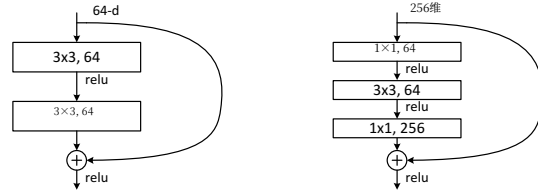


图 5. 用于 ImageNet 的更深的残差函数 $\mathcal{F}$ 。左：如 ResNet-34 的图 3 所示的一个构建块（在 56×56 特征图上）。右：用于 ResNet-50/101/152 的“瓶颈”构建块。

无参数的恒等捷径有助于训练。接下来我们研究投影捷径（式(2)）。在表3中，我们比较三种选项：（A）使用零填充捷径来增加维度，并且所有捷径都是无参数的（与表2和图4右侧相同）；（B）使用投影捷径来增加维度，其他捷径为恒等映射；以及（C）所有捷径都是投影。

表 3 表明，这三种选项都明显优于普通对应网络。B 略优于 A。我们认为，这是因为 A 中的零填充维度实际上没有进行残差学习。C 略优于 B，我们将此归因于许多（十三个）投影捷径引入的额外参数。但 A/B/C 之间的微小差异表明，投影捷径对于解决退化问题并非必需。因此，为降低内存/时间复杂度和模型大小，我们在本文其余部分不使用选项 C。恒等捷径对于不增加下文介绍的瓶颈架构的复杂度尤为重要。

更深的瓶颈架构。接下来我们描述用于 ImageNet 的更深网络。出于对可承受训练时间的考虑，我们将构建块修改为 **瓶颈** 设计⁴。对于每个残差函数 $\mathcal{F}$ ，我们使用 3 层堆叠而非 2 层（图 5）。这三个层分别是 1×1 、 3×3 和 1×1 卷积，其中 1×1 层负责降低维度，然后再增加（恢复）维度，从而使 3×3 层成为一个具有更小输入/输出维度的瓶颈。图 5 展示了一个示例，其中两种设计具有相似的时间复杂度。

无参数的恒等快捷连接对瓶颈架构尤为重要。如果将图 5（右）中的恒等快捷连接替换为投影，则可以证明，由于该快捷连接与两个高维端相连，其时间复杂度和模型大小都会加倍。因此，恒等快捷连接为瓶颈设计带来了更高效的模型。

50 层 ResNet: 我们将每个 2 层块在

⁴更深的非瓶颈 ResNet（例如，图 5 左）也能因深度增加而提高准确率（如 CIFAR-10 所示），但不如瓶颈 ResNet 经济。因此，使用瓶颈设计主要是出于实际考虑。我们进一步注意到，普通网络的退化问题在瓶颈设计中也同样存在。

34-layer net with this 3-layer bottleneck block, resulting in a 50-layer ResNet (Table 1). We use option B for increasing dimensions. This model has 3.8 billion FLOPs.

101-layer and 152-layer ResNets: We construct 101-layer and 152-layer ResNets by using more 3-layer blocks (Table 1). Remarkably, although the depth is significantly increased, the 152-layer ResNet (11.3 billion FLOPs) still has *lower complexity* than VGG-16/19 nets (15.3/19.6 billion FLOPs).

The 50/101/152-layer ResNets are more accurate than the 34-layer ones by considerable margins (Table 3 and 4). We do not observe the degradation problem and thus enjoy significant accuracy gains from considerably increased depth. The benefits of depth are witnessed for all evaluation metrics (Table 3 and 4).

Comparisons with State-of-the-art Methods. In Table 4 we compare with the previous best single-model results. Our baseline 34-layer ResNets have achieved very competitive accuracy. Our 152-layer ResNet has a single-model top-5 validation error of 4.49%. This single-model result outperforms all previous ensemble results (Table 5). We combine six models of different depth to form an ensemble (only with two 152-layer ones at the time of submitting). This leads to **3.57%** top-5 error on the test set (Table 5). *This entry won the 1st place in ILSVRC 2015.*

4.2. CIFAR-10 and Analysis

We conducted more studies on the CIFAR-10 dataset [20], which consists of 50k training images and 10k testing images in 10 classes. We present experiments trained on the training set and evaluated on the test set. Our focus is on the behaviors of extremely deep networks, but not on pushing the state-of-the-art results, so we intentionally use simple architectures as follows.

The plain/residual architectures follow the form in Fig. 3 (middle/right). The network inputs are 32×32 images, with the per-pixel mean subtracted. The first layer is 3×3 convolutions. Then we use a stack of $6n$ layers with 3×3 convolutions on the feature maps of sizes $\{32, 16, 8\}$ respectively, with $2n$ layers for each feature map size. The numbers of filters are $\{16, 32, 64\}$ respectively. The subsampling is performed by convolutions with a stride of 2. The network ends with a global average pooling, a 10-way fully-connected layer, and softmax. There are totally $6n+2$ stacked weighted layers. The following table summarizes the architecture:

output map size	32×32	16×16	8×8
# layers	$1+2n$	$2n$	$2n$
# filters	16	32	64

When shortcut connections are used, they are connected to the pairs of 3×3 layers (totally $3n$ shortcuts). On this dataset we use identity shortcuts in all cases (*i.e.*, option A),

method			error (%)
Maxout [10]			9.38
NIN [25]			8.81
DSN [24]			8.22
	# layers	# params	
FitNet [35]	19	2.5M	8.39
Highway [42, 43]	19	2.3M	7.54 (7.72±0.16)
Highway [42, 43]	32	1.25M	8.80
ResNet	20	0.27M	8.75
ResNet	32	0.46M	7.51
ResNet	44	0.66M	7.17
ResNet	56	0.85M	6.97
ResNet	110	1.7M	6.43 (6.61±0.16)
ResNet	1202	19.4M	7.93

Table 6. Classification error on the **CIFAR-10** test set. All methods are with data augmentation. For ResNet-110, we run it 5 times and show “best (mean±std)” as in [43].

so our residual models have exactly the same depth, width, and number of parameters as the plain counterparts.

We use a weight decay of 0.0001 and momentum of 0.9, and adopt the weight initialization in [13] and BN [16] but with no dropout. These models are trained with a mini-batch size of 128 on two GPUs. We start with a learning rate of 0.1, divide it by 10 at 32k and 48k iterations, and terminate training at 64k iterations, which is determined on a 45k/5k train/val split. We follow the simple data augmentation in [24] for training: 4 pixels are padded on each side, and a 32×32 crop is randomly sampled from the padded image or its horizontal flip. For testing, we only evaluate the single view of the original 32×32 image.

We compare $n = \{3, 5, 7, 9\}$, leading to 20, 32, 44, and 56-layer networks. Fig. 6 (left) shows the behaviors of the plain nets. The deep plain nets suffer from increased depth, and exhibit higher training error when going deeper. This phenomenon is similar to that on ImageNet (Fig. 4, left) and on MNIST (see [42]), suggesting that such an optimization difficulty is a fundamental problem.

Fig. 6 (middle) shows the behaviors of ResNets. Also similar to the ImageNet cases (Fig. 4, right), our ResNets manage to overcome the optimization difficulty and demonstrate accuracy gains when the depth increases.

We further explore $n = 18$ that leads to a 110-layer ResNet. In this case, we find that the initial learning rate of 0.1 is slightly too large to start converging⁵. So we use 0.01 to warm up the training until the training error is below 80% (about 400 iterations), and then go back to 0.1 and continue training. The rest of the learning schedule is as done previously. This 110-layer network converges well (Fig. 6, middle). It has *fewer* parameters than other deep and thin

⁵With an initial learning rate of 0.1, it starts converging (<90% error) after several epochs, but still reaches similar accuracy.

34 层网络中替换为这个 3 层瓶颈块，从而得到一个 50 层 ResNet（表 1）。我们使用选项 B 来增加维度。该模型有 38 亿 FLOPs。

101层和152层ResNet：我们通过使用更多的3层块来构建101层和152层的ResNet（表1）。值得注意的是，尽管深度显著增加，但152层ResNet（113亿FLOPs）仍然比VGG-16/19网络（153亿/196亿FLOPs）具有更低的复杂度。

50/101/152层ResNet比34层ResNet的准确率高很多（表3和表4）。我们没有观察到退化问题，因此从显著增加的深度中获得了显著的准确率提升。深度带来的好处在所有评估指标上都有体现（表3和表4）。

与最先进方法的比较。在表4中，我们与之前最佳的单模型结果进行了比较。我们的基线34层ResNet已经达到了非常有竞争力的准确率。我们的152层ResNet的单模型 top-5验证误差为4.49%。这个单模型结果超过了所有之前的集成结果（表5）。我们组合了六个不同深度的模型来形成一个集成（在提交时只用了两个152层的模型）。这在测试集上带来了**3.57%**的top-5误差（表5）。这个条目在ILSVRC 2015中赢得了第一名。

4.2. CIFAR-10 与分析

我们在 CIFAR-10 数据集[20], 上进行了更多研究, 该数据集包含 10 个类别中的 5 万张训练图像和 1 万张测试图像。我们展示了在训练集上训练并在测试集上评估的实验。我们关注的是极深网络的行为，而不是推动最先进的结果，因此我们有意使用如下简单架构。

普通/残差架构遵循图 3（中/右）的形式。网络输入是 32×32 图像，并减去逐像素均值。第一层是 3×3 卷积。然后我们使用一叠 $6n$ 层，在大小分别为 $\{32, 16, 8\}$ 的特征图上进行 3×3 卷积，其中每个特征图大小对应 $2n$ 层。滤波器数量分别为 $\{16, 32, 64\}$ 。下采样通过步长为 2 的卷积完成。网络以全局平均池化、10 路全连接层和 softmax 结尾。总共有 $6n+2$ 个堆叠的加权层。下表总结了该架构：

输出特征图大小	32×32	16×16	8×8
# 层数	$1+2n$	$2n$	$2n$
# 滤波器数	16	32	64

当使用快捷连接时，它们会连接到成对的 3×3 层（共 $3n$ 个快捷连接）。在此数据集上，我们在所有情况下都使用恒等快捷连接（即，选项 A），

方法	错误率 (%)		
Maxout [10]	9.38		
NIN [25]	8.81		
DSN [24]	8.22		
	#	# 参数	
FitNet [35]	19	2.5M	8.39
Highway [42, 43]	19	2.3M	7.54 (7.72±0.16)
Highway [42, 43]	32	1.25M	8.80
ResNet	20	0.27M	8.75
ResNet	32	0.46M	7.51
ResNet	44	0.66M	7.17
ResNet	56	0.85M	6.97
ResNet	110	1.7M	6.43 (6.61±0.16)
ResNet	1202	19.4M	7.93

表 6. **CIFAR-10** 测试集上的分类误差。所有方法均采用数据增强。对于 ResNet-110，我们运行 5 次，并展示“最佳（均值±标准差）”，如 [43] 所示。

因此，我们的残差模型与普通对应模型具有完全相同的深度、宽度，以及参数数量。

我们使用 0.0001 的权重衰减和 0.9 的动量，并采用 [13] 中的权重初始化以及 BN [16]，但不使用 dropout。这些模型使用 128 的小批量大小，在两块 GPU 上训练。我们以 0.1 的学习率开始，在 32k 和 48k 次迭代时将其除以 10，并在 64k 次迭代时终止训练，这一设置基于 45k/5k 的训练/验证集划分确定。我们按照 [24] 中用于训练的简单数据增强方法：每边填充 4 个像素，并从填充后的图像或其水平翻转中随机采样一个 32×32 裁剪区域。在测试时，我们仅评估原始 32×32 图像的单视图。

我们比较 $n = \{3, 5, 7, 9\}$ ，得到20、32、44和56层的网络。图6（左）展示了普通网络的行为。深层普通网络随着深度增加而出现退化，并且在更深时表现出更高的训练误差。这一现象与ImageNet（图4，左）和 MNIST（见 [42]）上的现象类似，表明这种优化困难是一个根本性问题。

图6（中）展示了ResNet的行为。同样与 ImageNet的情况（图4，右）类似，我们的ResNet设法克服了优化困难，并且在深度增加时表现出准确率提升。

我们进一步探索了导致 110 层 ResNet 的 $n = 18$ 。在这种情况下，我们发现初始学习率 0.1 略大，以致无法开始收敛⁵。因此，我们使用 0.01 预热训练，直到训练误差低于 80%（约 400 次迭代），然后再回到 0.1 并继续训练。其余学习计划与之前相同。这个 110 层网络收敛良好（图 6，中）。它的参数少于 其他深而窄的

⁵在初始学习率为0.1时，经过若干轮次后开始收敛 (<90%误差)，但仍达到相似的准确率。

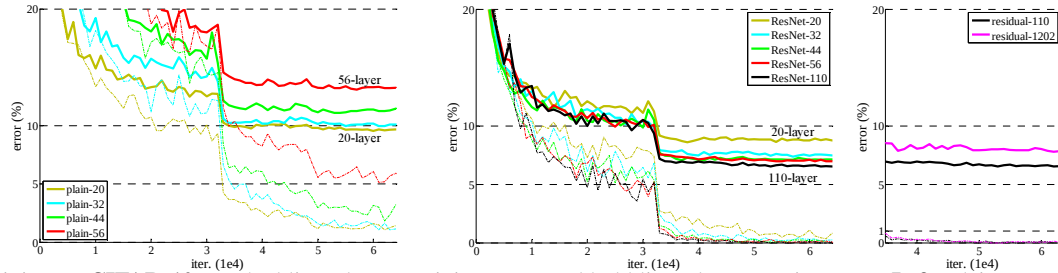


Figure 6. Training on **CIFAR-10**. Dashed lines denote training error, and bold lines denote testing error. **Left**: plain networks. The error of plain-110 is higher than 60% and not displayed. **Middle**: ResNets. **Right**: ResNets with 110 and 1202 layers.

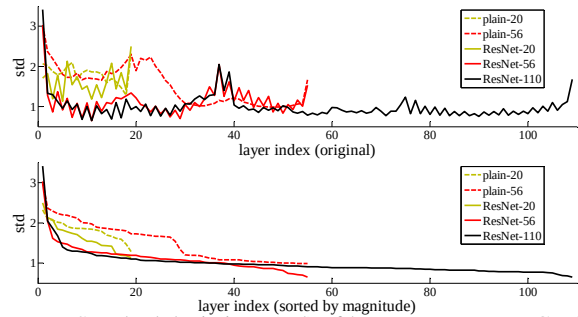


Figure 7. Standard deviations (std) of layer responses on CIFAR-10. The responses are the outputs of each 3×3 layer, after BN and before nonlinearity. **Top**: the layers are shown in their original order. **Bottom**: the responses are ranked in descending order.

networks such as FitNet [35] and Highway [42] (Table 6), yet is among the state-of-the-art results (6.43%, Table 6).

Analysis of Layer Responses. Fig. 7 shows the standard deviations (std) of the layer responses. The responses are the outputs of each 3×3 layer, after BN and before other nonlinearity (ReLU/addition). For ResNets, this analysis reveals the response strength of the residual functions. Fig. 7 shows that ResNets have generally smaller responses than their plain counterparts. These results support our basic motivation (Sec.3.1) that the residual functions might be generally closer to zero than the non-residual functions. We also notice that the deeper ResNet has smaller magnitudes of responses, as evidenced by the comparisons among ResNet-20, 56, and 110 in Fig. 7. When there are more layers, an individual layer of ResNets tends to modify the signal less.

Exploring Over 1000 layers. We explore an aggressively deep model of over 1000 layers. We set $n = 200$ that leads to a 1202-layer network, which is trained as described above. Our method shows *no optimization difficulty*, and this 10^3 -layer network is able to achieve *training error* $< 0.1\%$ (Fig. 6, right). Its test error is still fairly good (7.93%, Table 6).

But there are still open problems on such aggressively deep models. The testing result of this 1202-layer network is worse than that of our 110-layer network, although both

training data	07+12	07++12
test data	VOC 07 test	VOC 12 test
VGG-16	73.2	70.4
ResNet-101	76.4	73.8

Table 7. Object detection mAP (%) on the PASCAL VOC 2007/2012 test sets using **baseline** Faster R-CNN. See also Table 10 and 11 for better results.

metric	mAP@.5	mAP@[.5, .95]
VGG-16	41.5	21.2
ResNet-101	48.4	27.2

Table 8. Object detection mAP (%) on the COCO validation set using **baseline** Faster R-CNN. See also Table 9 for better results.

have similar training error. We argue that this is because of overfitting. The 1202-layer network may be unnecessarily large (19.4M) for this small dataset. Strong regularization such as maxout [10] or dropout [14] is applied to obtain the best results ([10, 25, 24, 35]) on this dataset. In this paper, we use no maxout/dropout and just simply impose regularization via deep and thin architectures by design, without distracting from the focus on the difficulties of optimization. But combining with stronger regularization may improve results, which we will study in the future.

4.3. Object Detection on PASCAL and MS COCO

Our method has good generalization performance on other recognition tasks. Table 7 and 8 show the object detection baseline results on PASCAL VOC 2007 and 2012 [5] and COCO [26]. We adopt *Faster R-CNN* [32] as the detection method. Here we are interested in the improvements of replacing VGG-16 [41] with ResNet-101. The detection implementation (see appendix) of using both models is the same, so the gains can only be attributed to better networks. Most remarkably, on the challenging COCO dataset we obtain a 6.0% increase in COCO’s standard metric (mAP@[.5, .95]), which is a 28% relative improvement. This gain is solely due to the learned representations.

Based on deep residual nets, we won the 1st places in several tracks in ILSVRC & COCO 2015 competitions: ImageNet detection, ImageNet localization, COCO detection, and COCO segmentation. The details are in the appendix.

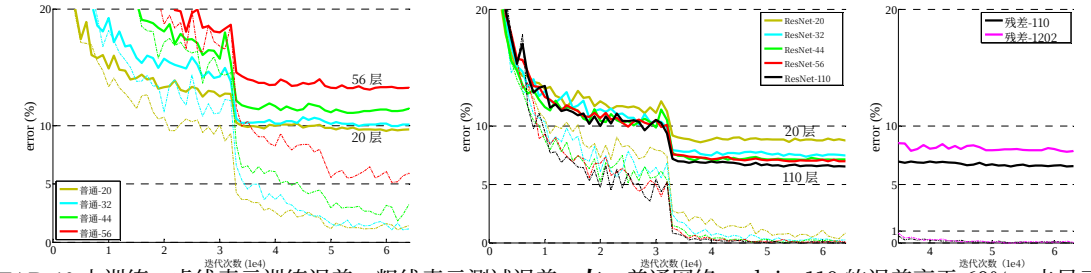


图 6. 在 **CIFAR-10** 上训练。虚线表示训练误差，粗线表示测试误差。左：普通网络。plain-110 的误差高于 60%，未显示。中：ResNets。右：具有 110 层和 1202 层的 ResNets。

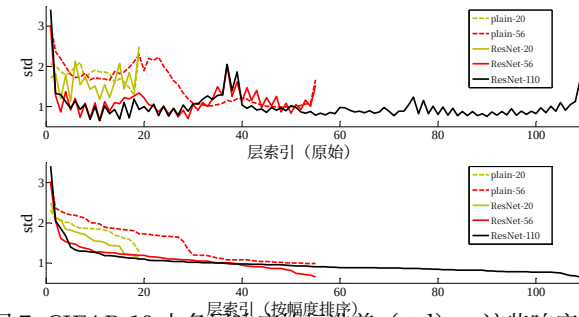


图 7. CIFAR-10 上各层响应的标准差 (std)。这些响应是每个 3×3 layer 在 BN 之后、非线性之前的输出。上：各层按原始顺序显示。下：响应按降序排列。

网络，如 FitNet [35] 和 Highway [42] (表 6)，却处于最先进结果之列 (6.43%，表 6)。

层响应分析。 图 7 显示了各层响应的标准差 (std)。这些响应是每个 3×3 layer 在 BN 之后、其他非线性 (ReLU/加法) 之前的输出。对于 ResNets，这一分析揭示了残差函数的响应强度。图 7 表明，ResNets 的响应通常比对应的普通网络更小。这些结果支持了我们的基本动机 (第 3.1 节)：残差函数通常可能比非残差函数更接近零。我们还注意到，更深的 ResNet 具有更小的响应幅度，图 7 中 ResNet-20、56 和 110 之间的比较证明了这一点。当层数更多时，ResNets 的单个层往往对信号的修改更少。

探索超过1000层。 我们探索了一个极其深的模型，超过 1000 层。我们设置 $n = 200$ ，从而得到一个 1202 层网络，并按照上述方式训练。我们的方法表明没有优化困难，而这个 10^3 层网络能够达到训练误差 $< 0.1\%$ (图 6, 右)。其测试误差仍然相当不错 (7.93%，表 6)。

但对于这种极其深的模型，仍然存在一些未解决的问题。这个 1202 层网络的测试结果比我们 110 层网络的测试结果更差，尽管二者

训练数据	07+12	07++12
测试数据	VOC 07 测试	VOC 12 测试
VGG-16	73.2	70.4
ResNet-101	76.4	73.8

表 7. 使用 **基线** Faster R-CNN 在 PASCAL VOC 2007/2012 测试集上的目标检测 mAP (%)。另见表 10 和 11 以获得更好的结果。

指标	mAP@.5	mAP@[.5, .95]
VGG-16	41.5	21.2
ResNet-101	48.4	27.2

表 8. 在 COCO 验证集上使用 **基线** Faster R-CNN 进行目标检测的 mAP (%)。另见表 9 以获得更好的结果。

具有相似的训练误差。我们认为这是因为过拟合。对于这个小数据集，1202 层网络可能没有必要如此大 (19.4M)。强正则化，例如 maxout [10] 或 dropout [14]，被用于在该数据集上获得最佳结果 ([10, 25, 24, 35])。在本文中，我们不使用 maxout/dropout，而只是通过设计深度且狭窄的架构来简单地施加正则化，而不偏离对优化难题的关注。不过，与更强的正则化相结合可能会改善结果，我们将在未来研究这一点。

4.3. 在 PASCAL 和 MS COCO 上的目标检测

我们的方法在其他识别任务上具有良好的泛化性能。表 7 和表 8 展示了在 PASCAL VOC 2007 和 2012 [5] 以及 COCO [26] 上的目标检测基线结果。我们采用 *Faster R-CNN* [32] 作为检测方法。这里我们感兴趣的是用 ResNet-101 替换 VGG-16 [41] 所带来的改进。使用两个模型的检测实现 (见附录) 是相同的，因此性能提升只能归因于更好的网络。最值得注意的是，在具有挑战性的 COCO 数据集上，我们在 COCO 的标准指标 (mAP@[.5, .95]) 上获得了 6.0% 的提升，即 28% 的相对提升。这一增益完全来自学习到的表示。

基于深度残差网络，我们在 ILSVRC 与 COCO 2015 竞赛的多个赛道中获得第一名：ImageNet 检测、ImageNet 定位、COCO 检测和 COCO 分割。详情见附录。

References

- [1] Y. Bengio, P. Simard, and P. Frasconi. Learning long-term dependencies with gradient descent is difficult. *IEEE Transactions on Neural Networks*, 5(2):157–166, 1994.
- [2] C. M. Bishop. *Neural networks for pattern recognition*. Oxford university press, 1995.
- [3] W. L. Briggs, S. F. McCormick, et al. *A Multigrid Tutorial*. Siam, 2000.
- [4] K. Chatfield, V. Lempitsky, A. Vedaldi, and A. Zisserman. The devil is in the details: an evaluation of recent feature encoding methods. In *BMVC*, 2011.
- [5] M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman. The Pascal Visual Object Classes (VOC) Challenge. *IJCV*, pages 303–338, 2010.
- [6] S. Gidaris and N. Komodakis. Object detection via a multi-region & semantic segmentation-aware cnn model. In *ICCV*, 2015.
- [7] R. Girshick. Fast R-CNN. In *ICCV*, 2015.
- [8] R. Girshick, J. Donahue, T. Darrell, and J. Malik. Rich feature hierarchies for accurate object detection and semantic segmentation. In *CVPR*, 2014.
- [9] X. Glorot and Y. Bengio. Understanding the difficulty of training deep feedforward neural networks. In *AISTATS*, 2010.
- [10] I. J. Goodfellow, D. Warde-Farley, M. Mirza, A. Courville, and Y. Bengio. Maxout networks. *arXiv:1302.4389*, 2013.
- [11] K. He and J. Sun. Convolutional neural networks at constrained time cost. In *CVPR*, 2015.
- [12] K. He, X. Zhang, S. Ren, and J. Sun. Spatial pyramid pooling in deep convolutional networks for visual recognition. In *ECCV*, 2014.
- [13] K. He, X. Zhang, S. Ren, and J. Sun. Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. In *ICCV*, 2015.
- [14] G. E. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. R. Salakhutdinov. Improving neural networks by preventing co-adaptation of feature detectors. *arXiv:1207.0580*, 2012.
- [15] S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural computation*, 9(8):1735–1780, 1997.
- [16] S. Ioffe and C. Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *ICML*, 2015.
- [17] H. Jegou, M. Douze, and C. Schmid. Product quantization for nearest neighbor search. *TPAMI*, 33, 2011.
- [18] H. Jegou, F. Perronnin, M. Douze, J. Sanchez, P. Perez, and C. Schmid. Aggregating local image descriptors into compact codes. *TPAMI*, 2012.
- [19] Y. Jia, E. Shelhamer, J. Donahue, S. Karayev, J. Long, R. Girshick, S. Guadarrama, and T. Darrell. Caffe: Convolutional architecture for fast feature embedding. *arXiv:1408.5093*, 2014.
- [20] A. Krizhevsky. Learning multiple layers of features from tiny images. *Tech Report*, 2009.
- [21] A. Krizhevsky, I. Sutskever, and G. Hinton. Imagenet classification with deep convolutional neural networks. In *NIPS*, 2012.
- [22] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel. Backpropagation applied to handwritten zip code recognition. *Neural computation*, 1989.
- [23] Y. LeCun, L. Bottou, G. B. Orr, and K.-R. Müller. Efficient backprop. In *Neural Networks: Tricks of the Trade*, pages 9–50. Springer, 1998.
- [24] C.-Y. Lee, S. Xie, P. Gallagher, Z. Zhang, and Z. Tu. Deeply-supervised nets. *arXiv:1409.5185*, 2014.
- [25] M. Lin, Q. Chen, and S. Yan. Network in network. *arXiv:1312.4400*, 2013.
- [26] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick. Microsoft COCO: Common objects in context. In *ECCV*. 2014.
- [27] J. Long, E. Shelhamer, and T. Darrell. Fully convolutional networks for semantic segmentation. In *CVPR*, 2015.

- [28] G. Montúfar, R. Pascanu, K. Cho, and Y. Bengio. On the number of linear regions of deep neural networks. In *NIPS*, 2014.
- [29] V. Nair and G. E. Hinton. Rectified linear units improve restricted boltzmann machines. In *ICML*, 2010.
- [30] F. Perronnin and C. Dance. Fisher kernels on visual vocabularies for image categorization. In *CVPR*, 2007.
- [31] T. Raiko, H. Valpola, and Y. LeCun. Deep learning made easier by linear transformations in perceptrons. In *AISTATS*, 2012.
- [32] S. Ren, K. He, R. Girshick, and J. Sun. Faster R-CNN: Towards real-time object detection with region proposal networks. In *NIPS*, 2015.
- [33] S. Ren, K. He, R. Girshick, X. Zhang, and J. Sun. Object detection networks on convolutional feature maps. *arXiv:1504.06066*, 2015.
- [34] B. D. Ripley. *Pattern recognition and neural networks*. Cambridge university press, 1996.
- [35] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio. Fitnets: Hints for thin deep nets. In *ICLR*, 2015.
- [36] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, et al. Imagenet large scale visual recognition challenge. *arXiv:1409.0575*, 2014.
- [37] A. M. Saxe, J. L. McClelland, and S. Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. *arXiv:1312.6120*, 2013.
- [38] N. N. Schraudolph. Accelerated gradient descent by factor-centering decomposition. Technical report, 1998.
- [39] N. N. Schraudolph. Centering neural network gradient factors. In *Neural Networks: Tricks of the Trade*, pages 207–226. Springer, 1998.
- [40] P. Sermanet, D. Eigen, X. Zhang, M. Mathieu, R. Fergus, and Y. LeCun. Overfeat: Integrated recognition, localization and detection using convolutional networks. In *ICLR*, 2014.
- [41] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. In *ICLR*, 2015.
- [42] R. K. Srivastava, K. Greff, and J. Schmidhuber. Highway networks. *arXiv:1505.00387*, 2015.
- [43] R. K. Srivastava, K. Greff, and J. Schmidhuber. Training very deep networks. *1507.06228*, 2015.
- [44] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich. Going deeper with convolutions. In *CVPR*, 2015.
- [45] R. Szeliski. Fast surface interpolation using hierarchical basis functions. *TPAMI*, 1990.
- [46] R. Szeliski. Locally adapted hierarchical basis preconditioning. In *SIGGRAPH*, 2006.
- [47] T. Vatanen, T. Raiko, H. Valpola, and Y. LeCun. Pushing stochastic gradient towards second-order methods—backpropagation learning with transformations in nonlinearities. In *Neural Information Processing*, 2013.
- [48] A. Vedaldi and B. Fulkerson. VLFeat: An open and portable library of computer vision algorithms, 2008.
- [49] W. Venables and B. Ripley. Modern applied statistics with s-plus. 1999.
- [50] M. D. Zeiler and R. Fergus. Visualizing and understanding convolutional neural networks. In *ECCV*, 2014.

参考文献

- [1] Y. Bengio、P. Simard 和 P. Frasconi. 用梯度下降学习长期依赖是困难的. 《IEEE神经网络汇刊》, 5(2):157–166, 1994.[2] C. M. Bishop. 用于模式识别的神经网络. 牛津大学出版社, 1995.[3] W. L. Briggs、S. F. McCormick 等. 多重网格教程. SIAM, 2000.
- [4] K. Chatfield、V. Lempitsky、A. Vedaldi 和 A. Zisserman. 细节决定成败：对近期特征编码方法的评估. 载于 BMVC, 2011.[5] M. Everingham、L. Van Gool、C. K. Williams、J. Winn 和 A. Zisserman. Pascal视觉对象类别（VOC）挑战赛. IJCV, 第303–338页, 2010.[6] S. Gidaris 和 N. Komodakis. 通过多区域与语义分割感知的 CNN 模型进行目标检测. 载于 ICCV, 2015.[7] R. Girshick. Fast R-CNN. 载于 ICCV, 2015.[8] R. Girshick、J. Donahue、T. Darrell 和 J. Malik. 用于精确目标检测和语义分割的丰富特征层级. 载于 CVPR, 2014.[9] X. Glorot 和 Y. Bengio. 理解训练深度前馈神经网络的困难. 载于 AISTATS, 2010.[10] I. J. Goodfellow、D. Warde-Farley、M. Mirza、A. Courville 和 Y. Bengio. Maxout 网络. arXiv:1302.4389, 2013.[11] K. He 和 J. Sun. 受限时间成本下的卷积神经网络. 载于 CVPR, 2015.[12] K. He、X. Zhang、S. Ren 和 J. Sun. 用于视觉识别的深度卷积网络中的空间金字塔池化. 载于 ECCV, 2014.[13] K. He、X. Zhang、S. Ren 和 J. Sun. 深入探究整流器：在 ImageNet 分类上超越人类水平表现. 载于 ICCV, 2015.[14] G. E. Hinton、N. Srivastava、A. Krizhevsky、I. Sutskever 和 R. R. Salakhutdinov. 通过防止特征检测器共适应来改进神经网络. arXiv:1207.0580, 2012.[15] S. Hochreiter 和 J. Schmidhuber. 长短期记忆. 《神经计算》, 9(8):1735–1780, 1997.[16] S. Ioffe 和 C. Szegedy. 批量归一化：通过减少内部协变量偏移来加速深度网络训练. 载于 ICML, 2015.[17] H. Jegou、M. Douze 和 C. Schmid. 用于最近邻搜索的乘积量化. TPAMI, 33, 2011.[18] H. Jegou、F. Perronnin、M. Douze、J. Sanchez、P. Perez 和 C. Schmid. 将局部图像描述子聚合为紧凑编码. TPAMI, 2012.[19] Y. Jia、E. Shelhamer、J. Donahue、S. Karayev、J. Long、R. Girshick、S. Guadarrama 和 T. Darrell. Caffe：用于快速特征嵌入的卷积架构. arXiv:1408.5093, 2014.[20] A. Krizhevsky. 从微小图像中学习多层特征. 技术报告, 2009.[21] A. Krizhevsky、I. Sutskever 和 G. Hinton. 使用深度卷积神经网络进行 ImageNet 分类. 载于 NIPS, 2012.[22] Y. LeCun、B. Boser、J. S. Denker、D. Henderson、R. E. Howard、W. Hubbard 和 L. D. Jackel. 应用于手写邮政编码识别的反向传播. 《神经计算》, 1989.[23] Y. LeCun、L. Bottou、G. B. Orr 和 K.-R. Müller. 高效反向传播. 载于《神经网络：行业诀窍》, 第9–50页. 施普林格, 1998.[24] C.-Y. Lee、S. Xie、P. Gallagher、Z. Zhang 和 Z. Tu. 深度监督网络. arXiv:1409.5185, 2014.[25] M. Lin、Q. Chen 和 S. Yan. 网络中的网络. arXiv:1312.4400, 2013.. [26] T.-Y. Lin、M. Maire、S. Belongie、J. Hays、P. Perona、D. Ramanan、P. Dollár 和 C. L. Zitnick. Microsoft COCO：上下文中的常见物体. 载于 ECCV. 2014.[27] J. Long、E. Shelhamer 和 T. Darrell. 用于语义分割的全卷积网络. 载于 CVPR, 2015.

- [28] G. Montúfar、R. Pascanu、K. Cho 和 Y. Bengio. 论深度神经网络的线性区域数量. 载于 NIPS, 2014年.[29] V. Nair 和 G. E. Hinton. 修正线性单元改进受限玻尔兹曼机. 载于 ICML, 2010年.[30] F. Perronnin 和 C. Dance. 用于图像分类的视觉词汇上的 Fisher 核. 载于 CVPR, 2007年.[31] T. Raiko、H. Valpola 和 Y. LeCun. 通过感知机中的线性变换使深度学习更容易. 载于 AISTATS, 2012年.[32] S. Ren、K. He、R. Girshick 和 J. Sun. Faster R-CNN：利用区域提议网络实现实时目标检测. 载于 NIPS, 2015年.[33] S. Ren、K. He、R. Girshick、X. Zhang 和 J. Sun. 基于卷积特征图的目标检测网络. arXiv:1504.06066, 2015年.[34] B. D. Ripley. 模式识别与神经网络. 剑桥大学出版社, 1996年.[35] A. Romero、N. Ballas、S. E. Kahou、A. Chassang、C. Gatta 和 Y. Bengio. FitNets：用于窄深网络的提示. 载于 ICLR, 2015年.[36] O. Russakovsky、J. Deng、H. Su、J. Krause、S. Satheesh、S. Ma、Z. Huang、A. Karpathy、A. Khosla、M. Bernstein 等. ImageNet 大规模视觉识别挑战赛. arXiv:1409.0575, 2014年.[37] A. M. Saxe、J. L. McClelland 和 S. Ganguli. 深度线性神经网络中学习非线性动力学的精确解. arXiv:1312.6120, 2013年.[38] N. N. Schraudolph. 通过因子中心化分解加速梯度下降. 技术报告, 1998年.[39] N. N. Schraudolph. 中心化神经网络梯度因子. 载于《神经网络：行家技巧》, 第 207–226 页. 斯普林格, 1998年.[40] P. Sermanet、D. Eigen、X. Zhang、M. Mathieu、R. Fergus 和 Y. LeCun. OverFeat：使用卷积网络进行集成识别、定位和检测. 载于 ICLR, 2014年.[41] K. Simonyan 和 A. Zisserman. 用于大规模图像识别的超深卷积网络. 载于 ICLR, 2015年.[42] R. K. Srivastava、K. Greff 和 J. Schmidhuber. 高速公路网络. arXiv:1505.00387, 2015年.[43] R. K. Srivastava、K. Greff 和 J. Schmidhuber. 训练极深网络. 1507.06228, 2015年.[44] C. Szegedy、W. Liu、Y. Jia、P. Sermanet、S. Reed、D. Anguelov、D. Erhan、V. Vanhoucke 和 A. Rabinovich. 用卷积走得更深. 载于 CVPR, 2015年.[45] R. Szeliski. 使用分层基函数的快速表面插值. TPAMI, 1990年.[46] R. Szeliski. 局部自适应的分层基预处理. 载于 SIGGRAPH, 2006年.[47] T. Vatanen、T. Raiko、H. Valpola 和 Y. LeCun. 将随机梯度推向二阶方法——在非线性中引入变换的反向传播学习. 载于《神经信息处理》, 2013年.[48] A. Vedaldi 和 B. Fulkerson. VLFeat：一个开放且可移植的计算机视觉算法库, 2008年.[49] W. Venables 和 B. Ripley. 使用 S-PLUS 的现代应用统计. 1999年.[50] M. D. Zeiler 和 R. Fergus. 可视化和理解卷积神经网络. 载于 ECCV, 2014年。

A. Object Detection Baselines

In this section we introduce our detection method based on the baseline Faster R-CNN [32] system. The models are initialized by the ImageNet classification models, and then fine-tuned on the object detection data. We have experimented with ResNet-50/101 at the time of the ILSVRC & COCO 2015 detection competitions.

Unlike VGG-16 used in [32], our ResNet has no hidden fc layers. We adopt the idea of “Networks on Conv feature maps” (NoC) [33] to address this issue. We compute the full-image shared conv feature maps using those layers whose strides on the image are no greater than 16 pixels (*i.e.*, conv1, conv2_x, conv3_x, and conv4_x, totally 91 conv layers in ResNet-101; Table 1). We consider these layers as analogous to the 13 conv layers in VGG-16, and by doing so, both ResNet and VGG-16 have conv feature maps of the same total stride (16 pixels). These layers are shared by a region proposal network (RPN, generating 300 proposals) [32] and a Fast R-CNN detection network [7]. RoI pooling [7] is performed before conv5_1. On this RoI-pooled feature, all layers of conv5_x and up are adopted for each region, playing the roles of VGG-16’s fc layers. The final classification layer is replaced by two sibling layers (classification and box regression [7]).

For the usage of BN layers, after pre-training, we compute the BN statistics (means and variances) for each layer on the ImageNet training set. Then the BN layers are fixed during fine-tuning for object detection. As such, the BN layers become linear activations with constant offsets and scales, and BN statistics are not updated by fine-tuning. We fix the BN layers mainly for reducing memory consumption in Faster R-CNN training.

PASCAL VOC

Following [7, 32], for the PASCAL VOC 2007 *test* set, we use the 5k *trainval* images in VOC 2007 and 16k *trainval* images in VOC 2012 for training (“07+12”). For the PASCAL VOC 2012 *test* set, we use the 10k *trainval+test* images in VOC 2007 and 16k *trainval* images in VOC 2012 for training (“07++12”). The hyper-parameters for training Faster R-CNN are the same as in [32]. Table 7 shows the results. ResNet-101 improves the mAP by >3% over VGG-16. This gain is solely because of the improved features learned by ResNet.

MS COCO

The MS COCO dataset [26] involves 80 object categories. We evaluate the PASCAL VOC metric (mAP @ IoU = 0.5) and the standard COCO metric (mAP @ IoU = .5:.05:.95). We use the 80k images on the train set for training and the 40k images on the val set for evaluation. Our detection system for COCO is similar to that for PASCAL VOC. We train the COCO models with an 8-GPU implementation, and thus the RPN step has a mini-batch size of

8 images (*i.e.*, 1 per GPU) and the Fast R-CNN step has a mini-batch size of 16 images. The RPN step and Fast R-CNN step are both trained for 240k iterations with a learning rate of 0.001 and then for 80k iterations with 0.0001.

Table 8 shows the results on the MS COCO validation set. ResNet-101 has a 6% increase of mAP@[.5, .95] over VGG-16, which is a 28% relative improvement, solely contributed by the features learned by the better network. Remarkably, the mAP@[.5, .95]’s absolute increase (6.0%) is nearly as big as mAP@.5’s (6.9%). This suggests that a deeper network can improve both recognition and localization.

B. Object Detection Improvements

For completeness, we report the improvements made for the competitions. These improvements are based on deep features and thus should benefit from residual learning.

MS COCO

Box refinement. Our box refinement partially follows the iterative localization in [6]. In Faster R-CNN, the final output is a regressed box that is different from its proposal box. So for inference, we pool a new feature from the regressed box and obtain a new classification score and a new regressed box. We combine these 300 new predictions with the original 300 predictions. Non-maximum suppression (NMS) is applied on the union set of predicted boxes using an IoU threshold of 0.3 [8], followed by box voting [6]. Box refinement improves mAP by about 2 points (Table 9).

Global context. We combine global context in the Fast R-CNN step. Given the full-image conv feature map, we pool a feature by global Spatial Pyramid Pooling [12] (with a “single-level” pyramid) which can be implemented as “RoI” pooling using the entire image’s bounding box as the RoI. This pooled feature is fed into the post-RoI layers to obtain a global context feature. This global feature is concatenated with the original per-region feature, followed by the sibling classification and box regression layers. This new structure is trained end-to-end. Global context improves mAP@.5 by about 1 point (Table 9).

Multi-scale testing. In the above, all results are obtained by single-scale training/testing as in [32], where the image’s shorter side is $s = 600$ pixels. Multi-scale training/testing has been developed in [12, 7] by selecting a scale from a feature pyramid, and in [33] by using maxout layers. In our current implementation, we have performed multi-scale *testing* following [33]; we have not performed multi-scale training because of limited time. In addition, we have performed multi-scale testing only for the Fast R-CNN step (but not yet for the RPN step). With a trained model, we compute conv feature maps on an image pyramid, where the image’s shorter sides are $s \in \{200, 400, 600, 800, 1000\}$.

A. 目标检测基线

在本节中，我们介绍基于基线 Faster R-CNN [32] 系统的检测方法。模型由 ImageNet 分类模型初始化，然后在目标检测数据上进行微调。在 ILSVRC 与 COCO 2015 检测竞赛期间，我们实验了 ResNet-50/101。

与 [32], 中使用的 VGG-16 不同，我们的 ResNet 没有隐藏的 fc 层。我们采用“卷积特征图上的网络”（Networks on Conv feature maps, NoC）的思想 [33] 来解决这个问题。我们使用那些在图像上的步长不超过 16 像素的层（即，conv1、conv2 x、conv3 x 和 conv4 x，在 ResNet-101 中总共 91 个卷积层；表 1）来计算整幅图像共享的卷积特征图。我们将这些层视为与 VGG-16 中的 13 个卷积层类似，这样一来，ResNet 和 VGG-16 都具有相同总步长（16 像素）的卷积特征图。这些层由区域提议网络（RPN，生成 300 个提议）[32] 和 Fast R-CNN 检测网络 [7] 共享。RoI 池化 [7] 在 conv5_1 之前执行。在这个 RoI 池化后的特征上，conv5_x 及以上的所有层都被用于每个区域，扮演 VGG-16 的 fc 层的角色。最终分类层被两个同级层（分类和边界框回归 [7]）替代。

关于 BN 层的使用，在预训练之后，我们在 ImageNet 训练集上计算每一层的 BN 统计量（均值和方差）。然后，在目标检测的微调过程中固定 BN 层。因此，BN 层变成具有恒定偏移和缩放比例的线性激活，并且 BN 统计量不会因微调而更新。我们固定 BN 层主要是为了减少 Faster R-CNN 训练中的内存消耗。

PASCAL VOC

按照 [7, 32], , 对于 PASCAL VOC 2007 测试集，我们使用 VOC 2007 中的 5k 训练验证图像和 VOC 2012 中的 16k 训练验证图像进行训练（“07+12”）。对于 PASCAL VOC 2012 测试集，我们使用 VOC 2007 中的 10k 训练验证+测试图像以及 VOC 2012 中的 16k 训练验证图像进行训练（“07++12”）。训练 Faster R-CNN 的超参数与 [32] 中相同。表 7 展示了结果。ResNet-101 相比 VGG-16 将 mAP 提高了 >3%。这一提升完全来自 ResNet 学习到的改进特征。

MS COCO

MS COCO 数据集 [26] 包含 80 个对象类别。我们评估 PASCAL VOC 指标（mAP@IoU = 0.5）和标准 COCO 指标（mAP @IoU =.5:.05:.95）。我们使用训练集上的 8 万张图像进行训练，并使用验证集上的 4 万张图像进行评估。我们的 COCO 检测系统与 PASCAL VOC 的检测系统类似。我们采用 8 个 GPU 的实现训练 COCO 模型，因此 RPN 步骤的小批量大小为

8 张图像（即，每块 GPU 1 张），而 Fast R-CNN 步骤的小批量大小为 16 张图像。RPN 步骤和 Fast R-CNN 步骤都先以 0.001 的学习率训练 24 万次迭代，然后再以 0.0001 的学习率训练 8 万次迭代。

表 8 展示了在 MS COCO 验证集上的结果。ResNet-101 的 mAP@[.5, .95] 相比 VGG-16 提高了 6%，相对提升为 28%，这完全归功于更好的网络所学习到的特征。值得注意的是，mAP@[.5, .95] 的绝对提升（6.0%）几乎与 mAP@.5 的（6.9%）一样大。这表明更深的网络可以同时改善识别和定位。

B. 目标检测改进

为完整起见，我们报告为竞赛所做的改进。这些改进基于深度特征，因此应当能从残差学习中受益。

MS COCO

边界框细化。我们的边界框细化部分遵循了 [6] 中的迭代定位。在 Faster R-CNN 中，最终输出是一个回归框，它不同于其候选框。因此，在推理时，我们从回归框中池化一个新特征，并获得新的分类得分和新的回归框。我们将这 300 个新预测与原来的 300 个预测结合起来。使用 IoU 阈值 0.3 [8]，对预测框的并集应用非极大值抑制（NMS），随后进行边界框投票 [6]。边界框细化将 mAP 提高了约 2 个点（表 9）。

全局上下文。我们在 Fast R-CNN 步骤中结合全局上下文。给定整幅图像的卷积特征图，我们使用全局空间金字塔池化 [12]（采用“单层”金字塔）池化一个特征，这可以通过将整幅图像的边界框作为 RoI 的“RoI”池化来实现。该池化特征被送入 RoI 后的各层，以获得全局上下文特征。该全局特征与原始的逐区域特征拼接，其后接同级的分类层和边界框回归层。这个新结构以端到端方式训练。全局上下文将 mAP@.5 提高约 1 个百分点（表 9）。

多尺度测试。在上述内容中，所有结果均按照 [32], 中的单尺度训练/测试获得，其中图像的较短边为 $s = 600$ 像素。多尺度训练/测试已在 [12, 7] 中通过从特征金字塔中选择一个尺度得到发展，并在 [33] 中通过使用 maxout 层得到发展。在我们当前的实现中，我们按照 [33] 进行了多尺度测试；由于时间有限，我们尚未进行多尺度训练。此外，我们仅对 Fast R-CNN 步骤进行了多尺度测试（但尚未对 RPN 步骤进行）。对于训练好的模型，我们在图像金字塔上计算卷积特征图，其中图像的较短边为 $s \in \{200, 400, 600, 800, 1000\}$ 。

training data	COCO train		COCO trainval	
test data	COCO val		COCO test-dev	
mAP	@.5	@[.5, .95]	@.5	@[.5, .95]
baseline Faster R-CNN (VGG-16)	41.5	21.2		
baseline Faster R-CNN (ResNet-101)	48.4	27.2		
+box refinement	49.9	29.9		
+context	51.1	30.0	53.3	32.2
+multi-scale testing	53.8	32.5	55.7	34.9
ensemble			59.0	37.4

Table 9. Object detection improvements on MS COCO using Faster R-CNN and ResNet-101.

system	net	data	mAP	areo	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv
baseline	VGG-16	07+12	73.2	76.5	79.0	70.9	65.5	52.1	83.1	84.7	86.4	52.0	81.9	65.7	84.8	84.6	77.5	76.7	38.8	73.6	73.9	83.0	72.6
baseline	ResNet-101	07+12	76.4	79.8	80.7	76.2	68.3	55.9	85.1	85.3	89.8	56.7	87.8	69.4	88.3	88.9	80.9	78.4	41.7	78.6	79.8	85.3	72.0
baseline+++	ResNet-101	COCO+07+12	85.6	90.0	89.6	87.8	80.8	76.1	89.9	89.9	89.6	75.5	90.0	80.7	89.6	90.3	89.1	88.7	65.4	88.1	85.6	89.0	86.8

Table 10. Detection results on the PASCAL VOC 2007 test set. The baseline is the Faster R-CNN system. The system “baseline+++” include box refinement, context, and multi-scale testing in Table 9.

system	net	data	mAP	areo	bike	bird	boat	bottle	bus	car	cat	chair	cow	table	dog	horse	mbike	person	plant	sheep	sofa	train	tv
baseline	VGG-16	07++12	70.4	84.9	79.8	74.3	53.9	49.8	77.5	75.9	88.5	45.6	77.1	55.3	86.9	81.7	80.9	79.6	40.1	72.6	60.9	81.2	61.5
baseline	ResNet-101	07++12	73.8	86.5	81.6	77.2	58.0	51.0	78.6	76.6	93.2	48.6	80.4	59.0	92.1	85.3	84.8	80.7	48.1	77.3	66.5	84.7	65.6
baseline+++	ResNet-101	COCO+07++12	83.8	92.1	88.4	84.8	75.9	71.4	86.3	87.8	94.2	66.8	89.4	69.2	93.9	91.9	90.9	89.6	67.9	88.2	76.8	90.3	80.0

Table 11. Detection results on the PASCAL VOC 2012 test set (<http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?challengeid=11&compid=4>). The baseline is the Faster R-CNN system. The system “baseline+++” include box refinement, context, and multi-scale testing in Table 9.

We select two adjacent scales from the pyramid following [33]. RoI pooling and subsequent layers are performed on the feature maps of these two scales [33], which are merged by maxout as in [33]. Multi-scale testing improves the mAP by over 2 points (Table 9).

Using validation data. Next we use the 80k+40k trainval set for training and the 20k test-dev set for evaluation. The test-dev set has no publicly available ground truth and the result is reported by the evaluation server. Under this setting, the results are an mAP@.5 of 55.7% and an mAP@[.5, .95] of 34.9% (Table 9). This is our single-model result.

Ensemble. In Faster R-CNN, the system is designed to learn region proposals and also object classifiers, so an ensemble can be used to boost both tasks. We use an ensemble for proposing regions, and the union set of proposals are processed by an ensemble of per-region classifiers. Table 9 shows our result based on an ensemble of 3 networks. The mAP is 59.0% and 37.4% on the test-dev set. *This result won the 1st place in the detection task in COCO 2015.*

PASCAL VOC

We revisit the PASCAL VOC dataset based on the above model. With the single model on the COCO dataset (55.7% mAP@.5 in Table 9), we fine-tune this model on the PASCAL VOC sets. The improvements of box refinement, context, and multi-scale testing are also adopted. By doing so

	val2	test
GoogLeNet [44] (ILSVRC’ 14)	-	43.9
our single model (ILSVRC’ 15)	60.5	58.8
our ensemble (ILSVRC’ 15)	63.6	62.1

Table 12. Our results (mAP, %) on the ImageNet detection dataset. Our detection system is Faster R-CNN [32] with the improvements in Table 9, using ResNet-101.

we achieve 85.6% mAP on PASCAL VOC 2007 (Table 10) and 83.8% on PASCAL VOC 2012 (Table 11)⁶. The result on PASCAL VOC 2012 is 10 points higher than the previous state-of-the-art result [6].

ImageNet Detection

The ImageNet Detection (DET) task involves 200 object categories. The accuracy is evaluated by mAP@.5. Our object detection algorithm for ImageNet DET is the same as that for MS COCO in Table 9. The networks are pre-trained on the 1000-class ImageNet classification set, and are fine-tuned on the DET data. We split the validation set into two parts (val1/val2) following [8]. We fine-tune the detection models using the DET training set and the val1 set. The val2 set is used for validation. We do not use other ILSVRC 2015 data. Our single model with ResNet-101 has

⁶<http://host.robots.ox.ac.uk:8080/anonymous/30J40J.html>, submitted on 2015-11-26.

训练数据	COCO 训练集		COCO 训练验证集	
测试数据	COCO 验证集		COCO 测试开发集	
mAP	@.5	@[.5, .95]	@.5	@[.5, .95]
基线 Faster R-CNN (VGG-16)	41.5	21.2		
基线 Faster R-CNN (ResNet-101)	48.4	27.2		
+边界框细化	49.9	29.9		
+上下文	51.1	30.0	53.3	32.2
+多尺度测试	53.8	32.5	55.7	34.9
集成			59.0	37.4

表 9. 在 MS COCO 上使用 Faster R-CNN 和 ResNet-101 对目标检测的改进。

系统	net	data	mAP	areo	bike	bird	boat	瓶子	bus	car	cat	椅子	cow	桌子	dog	马	摩托车	人	植物	羊	sofa	火车	tv
基线	VGG-16	07+12	73.2	76.5	79.0	70.9	65.5	52.1	83.1	84.7	86.4	52.0	81.9	65.7	84.8	84.6	77.5	76.7	38.8	73.6	73.9	83.0	72.6
基线	ResNet-101	07+12	76.4	79.8	80.7	76.2	68.3	55.9	85.1	85.3	89.8	56.7	87.8	69.4	88.3	88.9	80.9	78.4	41.7	78.6	79.8	85.3	72.0
基线+++	ResNet-101	COCO+07+12	85.6	90.0	89.6	87.8	80.8	76.1	89.9	89.9	89.6	75.5	90.0	80.7	89.6	90.3	89.1	88.7	65.4	88.1	85.6	89.0	86.8

表 10. PASCAL VOC 2007 测试集上的检测结果。基线是 Faster R-CNN 系统。系统 “baseline+++” 包括边框细化、上下文和多尺度测试，见表 9。

系统	net	data	mAP	areo	bike	bird	boat	瓶子	bus	car	cat	椅子	cow	桌子	dog	马	摩托车	人	植物	羊	sofa	火车	tv
基线	VGG-16	07++12	70.4	84.9	79.8	74.3	53.9	49.8	77.5	75.9	88.5	45.6	77.1	55.3	86.9	81.7	80.9	79.6	40.1	72.6	60.9	81.2	61.5
基线	ResNet-101	07++12	73.8	86.5	81.6	77.2	58.0	51.0	78.6	76.6	93.2	48.6	80.4	59.0	92.1	85.3	84.8	80.7	48.1	77.3	66.5	84.7	65.6
基线+++	ResNet-101	COCO+07++12	83.8	92.1	88.4	84.8	75.9	71.4	86.3	87.8	94.2	66.8	89.4	69.2	93.9	91.9	90.9	89.6	67.9	88.2	76.8	90.3	80.0

表 11. PASCAL VOC 2012 测试集上的检测结果（<http://host.robots.ox.ac.uk:8080/leaderboard/displaylb.php?challengeid=11&compid=4>）。基线是 Faster R-CNN 系统。系统 “基线+++” 在表 9 中包括边界框细化、上下文和多尺度测试。

我们按照 [33] 从金字塔中选择两个相邻尺度。RoI 池化和后续层在这两个尺度的特征图 [33], 上执行，这些特征图按照 [33] 中的方式通过 maxout 合并。多尺度测试将 mAP 提升了超过 2 个点（表 9）。

使用验证数据。接下来，我们使用 80k+40k 的 trainval 集进行训练，并使用 20k 的 test-dev 集进行评估。test-dev 集没有公开可用的真实标注，结果由评估服务器报告。在此设置下，mAP@.5 为 55.7%，mAP@[.5, .95] 为 34.9%（表 9）。这是我们单模型的结果。

集成。在 Faster R-CNN 中，系统被设计用于学习区域提议和目标分类器，因此可以使用集成来提升这两项任务。我们使用集成来生成区域提议，而提议的并集由逐区域分类器的集成进行处理。表 9 展示了我们基于 3 个网络集成得到的结果。在 test-dev 集上，mAP 分别为 59.0% 和 37.4%。该结果在 *COCO 2015* 检测任务中荣获第一名。

PASCAL VOC

我们基于上述模型重新审视 PASCAL VOC 数据集。利用在 COCO 数据集上的单模型（表 9 中 mAP@.5 为 55.7%），我们在 PASCAL VOC 数据集上对该模型进行微调。还采用了边界框精修、上下文和多尺度测试等改进。如此一来，

	val2	test
GoogLeNet [44] (ILSVRC’ 14)	-	43.9
我们的单模型（ILSVRC’ 15）	60.5	58.8
我们的集成模型（ILSVRC’ 15）	63.6	62.1

表12. 我们在 ImageNet 检测数据集上的结果（mAP，%）。我们的检测系统是 Faster R-CNN [32]，采用了表9中的改进，使用 ResNet-101。

我们在 PASCAL VOC 2007 上达到了 85.6% mAP（表 10），在 PASCAL VOC 2012 上达到 83.8%（表 11）⁶。在 PASCAL VOC 2012 上的结果比此前的最先进结果 [6] 高出 10 个百分点。

ImageNet 检测

ImageNet 检测（DET）任务涉及 200 个目标类别。准确率通过 mAP@.5 评估。我们的 ImageNet DET 目标检测算法与表 9 中用于 MS COCO 的算法相同。这些网络在 1000 类 ImageNet 分类集上预训练，并在 DET 数据上微调。我们遵循 [8] 将验证集划分为两部分（val1/val2）。我们使用 DET 训练集和 val1 集微调检测模型。val2 集用于验证。我们不使用其他 ILSVRC 2015 数据。我们采用 ResNet-101 的单个模型取得了

⁶<http://host.robots.ox.ac.uk:8080/anonymous/30J40J.html>, 提交于 2015-11-26。

LOC method	LOC network	testing	LOC error on GT CLS	classification network	top-5 LOC error on predicted CLS
VGG’s [41]	VGG-16	1-crop	33.1 [41]		
RPN	ResNet-101	1-crop	13.3		
RPN	ResNet-101	dense	11.7		
RPN	ResNet-101	dense		ResNet-101	14.4
RPN+RCNN	ResNet-101	dense		ResNet-101	10.6
RPN+RCNN	ensemble	dense		ensemble	8.9

Table 13. Localization error (%) on the ImageNet validation. In the column of “LOC error on GT class” ([41]), the ground truth class is used. In the “testing” column, “1-crop” denotes testing on a center crop of 224×224 pixels, “dense” denotes dense (fully convolutional) and multi-scale testing.

58.8% mAP and our ensemble of 3 models has 62.1% mAP on the DET test set (Table 12). *This result won the 1st place in the ImageNet detection task in ILSVRC 2015*, surpassing the second place by **8.5 points** (absolute).

C. ImageNet Localization

The ImageNet Localization (LOC) task [36] requires to classify and localize the objects. Following [40, 41], we assume that the image-level classifiers are first adopted for predicting the class labels of an image, and the localization algorithm only accounts for predicting bounding boxes based on the predicted classes. We adopt the “per-class regression” (PCR) strategy [40, 41], learning a bounding box regressor for each class. We pre-train the networks for ImageNet classification and then fine-tune them for localization. We train networks on the provided 1000-class ImageNet training set.

Our localization algorithm is based on the RPN framework of [32] with a few modifications. Unlike the way in [32] that is category-agnostic, our RPN for localization is designed in a *per-class* form. This RPN ends with two sibling 1×1 convolutional layers for binary classification (*cls*) and box regression (*reg*), as in [32]. The *cls* and *reg* layers are both in a *per-class* from, in contrast to [32]. Specifically, the *cls* layer has a 1000-d output, and each dimension is *binary logistic regression* for predicting being or not being an object class; the *reg* layer has a 1000×4-d output consisting of box regressors for 1000 classes. As in [32], our bounding box regression is with reference to multiple translation-invariant “anchor” boxes at each position.

As in our ImageNet classification training (Sec. 3.4), we randomly sample 224×224 crops for data augmentation. We use a mini-batch size of 256 images for fine-tuning. To avoid negative samples being dominate, 8 anchors are randomly sampled for each image, where the sampled positive and negative anchors have a ratio of 1:1 [32]. For testing, the network is applied on the image fully-convolutionally.

Table 13 compares the localization results. Following [41], we first perform “oracle” testing using the ground truth class as the classification prediction. VGG’s paper [41] re-

method	top-5 localization err	
	val	test
OverFeat [40] (ILSVRC’ 13)	30.0	29.9
GoogLeNet [44] (ILSVRC’ 14)	-	26.7
VGG [41] (ILSVRC’ 14)	26.9	25.3
ours (ILSVRC’ 15)	8.9	9.0

Table 14. Comparisons of localization error (%) on the ImageNet dataset with state-of-the-art methods.

ports a center-crop error of 33.1% (Table 13) using ground truth classes. Under the same setting, our RPN method using ResNet-101 net significantly reduces the center-crop error to 13.3%. This comparison demonstrates the excellent performance of our framework. With dense (fully convolutional) and multi-scale testing, our ResNet-101 has an error of 11.7% using ground truth classes. Using ResNet-101 for predicting classes (4.6% top-5 classification error, Table 4), the top-5 localization error is 14.4%.

The above results are only based on the *proposal network* (RPN) in Faster R-CNN [32]. One may use the *detection network* (Fast R-CNN [7]) in Faster R-CNN to improve the results. But we notice that on this dataset, one image usually contains a single dominate object, and the proposal regions highly overlap with each other and thus have very similar RoI-pooled features. As a result, the image-centric training of Fast R-CNN [7] generates samples of small variations, which may not be desired for stochastic training. Motivated by this, in our current experiment we use the original R-CNN [8] that is RoI-centric, in place of Fast R-CNN.

Our R-CNN implementation is as follows. We apply the per-class RPN trained as above on the training images to predict bounding boxes for the ground truth class. These predicted boxes play a role of class-dependent proposals. For each training image, the highest scored 200 proposals are extracted as training samples to train an R-CNN classifier. The image region is cropped from a proposal, warped to 224×224 pixels, and fed into the classification network as in R-CNN [8]. The outputs of this network consist of two sibling fc layers for *cls* and *reg*, also in a per-class form. This R-CNN network is fine-tuned on the training set using a mini-batch size of 256 in the RoI-centric fashion. For testing, the RPN generates the highest scored 200 proposals for each predicted class, and the R-CNN network is used to update these proposals’ scores and box positions.

This method reduces the top-5 localization error to 10.6% (Table 13). This is our single-model result on the validation set. Using an ensemble of networks for both classification and localization, we achieve a top-5 localization error of 9.0% on the test set. This number significantly outperforms the ILSVRC 14 results (Table 14), showing a 64% relative reduction of error. *This result won the 1st place in the ImageNet localization task in ILSVRC 2015*.

LOC method	LOC network	testing	定位误差在 GT CLS 上	分类网络	top-5 LOC 误差在预测的 CLS 上
VGG 的 [41]	VGG-16	单裁剪	33.1 [41]		
RPN	ResNet-101	单裁剪	13.3		
RPN	ResNet-101	密集	11.7		
RPN	ResNet-101	密集		ResNet-101	14.4
RPN+RCNN	ResNet-101	密集		ResNet-101	10.6
RPN+RCNN	集成	稠密		集成	8.9

表 13. ImageNet 验证集上的定位误差 (%)。在 “GT 类的 LOC 误差” （[41]）这一列中，使用真实类别。在 “测试” 列中，“1-crop” 表示在 224×224 像素的中心裁剪图像上进行测试，“dense” 表示稠密（全卷积）和多尺度测试。

58.8% mAP，而我们 3 个模型的集成在 DET 测试集上达到 62.1% mAP（表 12）。这一结果在 *ILSVRC 2015* 的 *ImageNet* 检测任务中获得第 1 名，以8.5个百分点（绝对值）超过第二名。

C. ImageNet 定位

ImageNet 定位（LOC）任务 [36] 要求对物体进行分类和定位。按照 [40, 41], , 我们假设首先采用图像级分类器来预测图像的类别标签，而定位算法仅负责根据预测出的类别来预测边界框。我们采用“逐类回归”（PCR）策略 [40, 41], 为每个类别学习一个边界框回归器。我们为 ImageNet 分类预训练网络，然后针对定位任务对其进行微调。我们在所提供的 1000 类 ImageNet 训练集上训练网络。

我们的定位算法基于 [32] 的 RPN 框架，并做了一些修改。与 [32] 中类别无关的方式不同，我们用于定位的 RPN 以逐类形式设计。该 RPN 以两个并列的 1×1 卷积层结束，分别用于二分类（*cls*）和边界框回归（*reg*），如 [32]中那样。*cls* 和 *reg* 层也都采用逐类形式，与 [32]相反。具体来说，*cls*层有 1000 维输出，每一维都是 二元逻辑回归，用于预测是否为某一物体类别；*reg* 层有 1000×4 维输出，由 1000 个类别的边界框回归器组成。如 [32], 中那样，我们的边界框回归参考每个位置上的多个平移不变“锚”框。

正如我们在ImageNet分类训练（第3.4节）中那样，我们随机采样 224×224 个裁剪区域进行数据增强。我们使用256张图像的小批量大小进行微调。为了避免负样本占主导地位，每张图像随机采样8个锚框，其中采样的正负锚框的比例为1:1 [32]。在测试时，网络以全卷积方式应用于图像。

表13比较了定位结果。参照[41], , 我们首先使用真实类别作为分类预测进行“oracle”测试。VGG的论文 [41] 报

方法	top-5定位误差	
	val	test
OverFeat [40] (ILSVRC’ 13)	30.0	29.9
GoogLeNet [44] (ILSVRC’ 14)	-	26.7
VGG [41](ILSVRC’ 14)	26.9	25.3
我们的方法 (ILSVRC’ 15)	8.9	9.0

表 14. 在 ImageNet 数据集上与最先进方法的定位误差 (%) 比较。

告了在使用真实类别时，中心裁剪误差为33.1%（表13）。在相同设置下，我们采用ResNet-101网络的RPN方法将中心裁剪误差显著降低至13.3%。这一比较展示了我们框架的出色性能。在密集（全卷积）和多尺度测试下，我们的ResNet-101在使用真实类别时的误差为11.7%。使用ResNet-101进行类别预测（4.6%的top-5分类误差，表4）时，top-5定位误差为14.4%。

上述结果仅基于 Faster R-CNN [32]中的提议网络（RPN）。可以使用 Faster R-CNN 中的检测网络（即 Fast R-CNN [7]）来改进结果。但我们注意到，在这个数据集上，一幅图像通常只包含一个占主导地位的物体，而提议区域彼此高度重叠，因此具有非常相似的RoI 池化特征。因此，Fast R-CNN [7]的以图像为中心的训练会生成变化很小的样本，这可能不是随机训练所需要的。受此启发，在我们当前的实验中，我们使用原始的 R-CNN [8], 它是以 RoI 为中心的，来代替 Fast R-CNN。

我们的 R-CNN 实现如下。我们在训练图像上应用如上训练的逐类 RPN，以预测真实类别的边界框。这些预测框起到类别相关提议的作用。对于每张训练图像，提取得分最高的 200 个提议作为训练样本来训练一个 R-CNN 分类器。图像区域从提议中裁剪出来，缩放至 224×224像素，并像 R-CNN [8]中那样输入分类网络。该网络的输出由两个兄弟 fc 层组成，分别用于分类和回归，同样采用逐类形式。该 R-CNN 网络以 RoI 为中心的方式，在训练集上使用 256 的小批量大小进行微调。对于测试，RPN 为每个预测类别生成得分最高的 200 个提议，并使用 R-CNN 网络来更新这些提议的得分和框位置。

该方法将 top-5 定位误差降低至 10.6%（表 13）。这是我们在验证集上的单模型结果。对分类和定位均使用网络集成后，我们在测试集上取得了 9.0% 的 top-5 定位误差。这一数字显著优于 ILSVRC 14 的结果（表 14），相对误差降低了 64%。该结果在 *ILSVRC 2015* 的 *ImageNet* 定位任务中赢得了第一名。